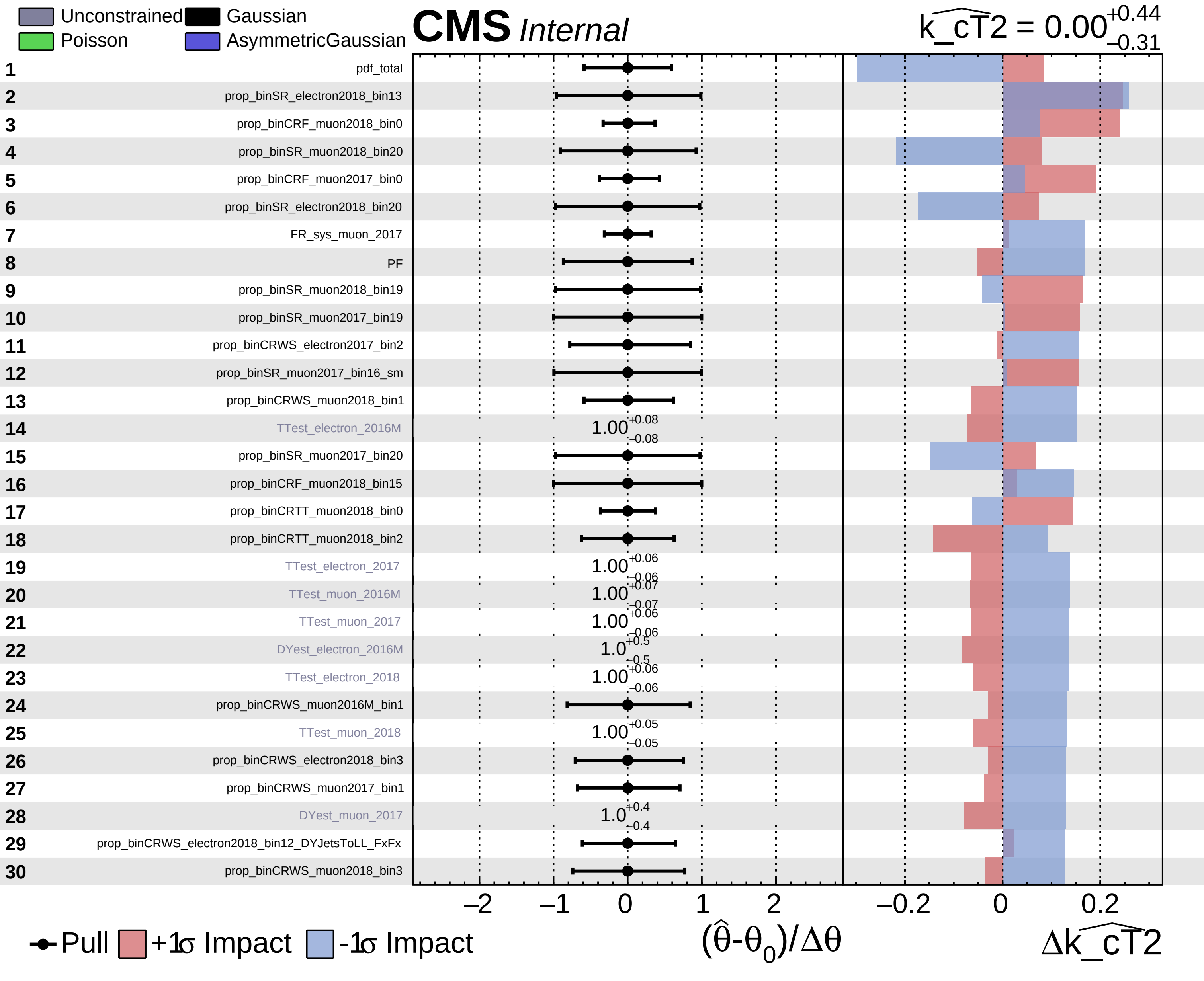
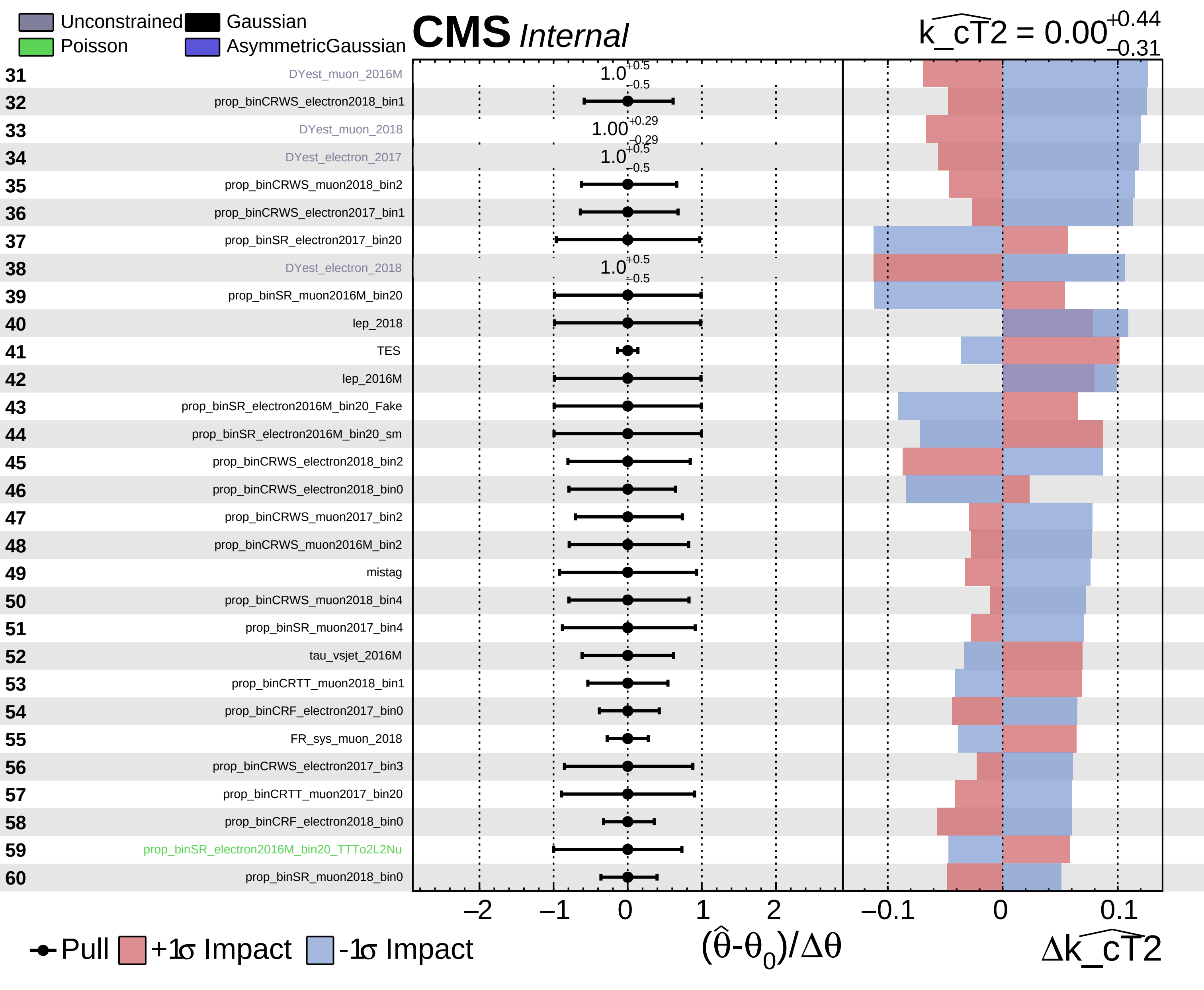
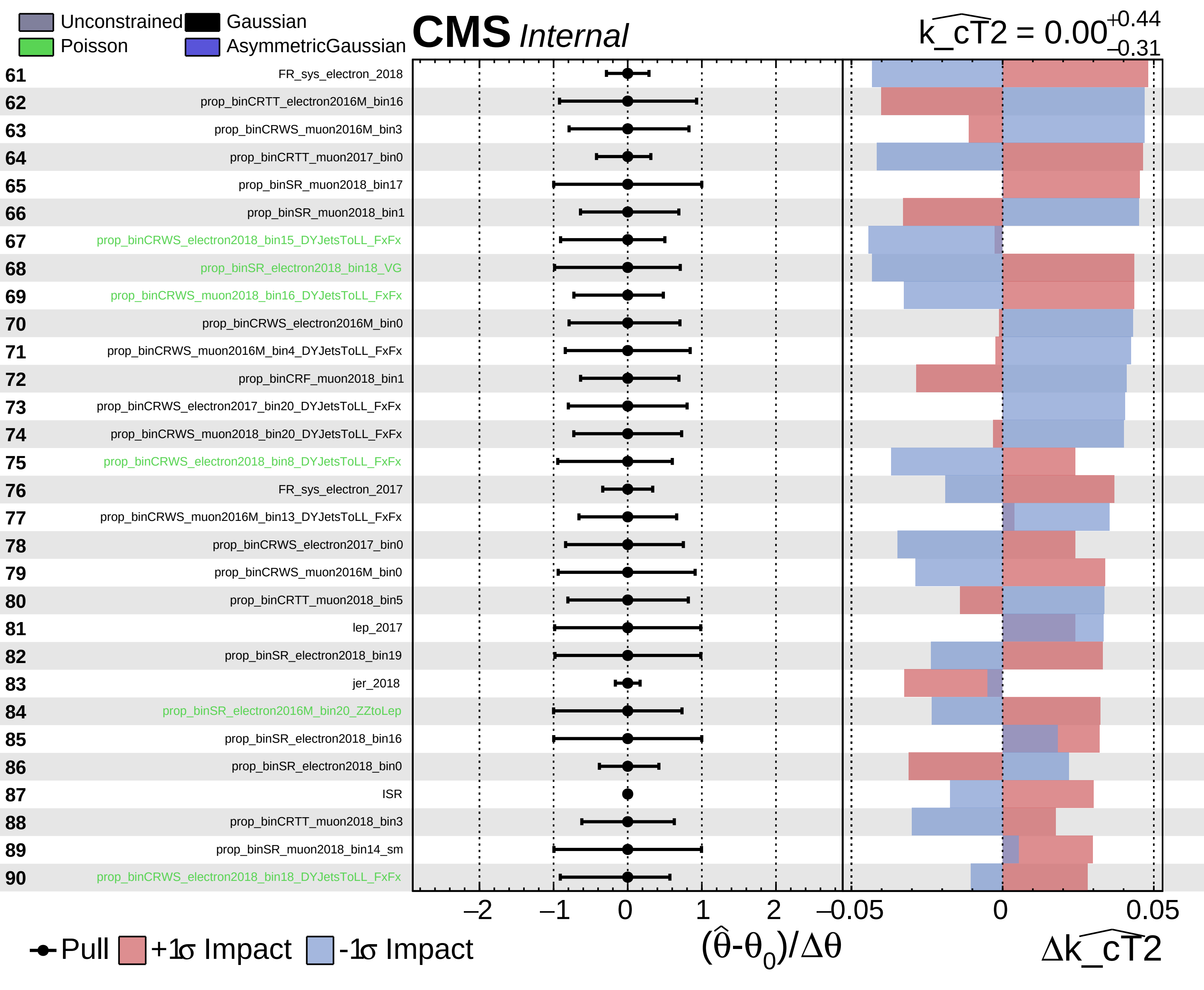
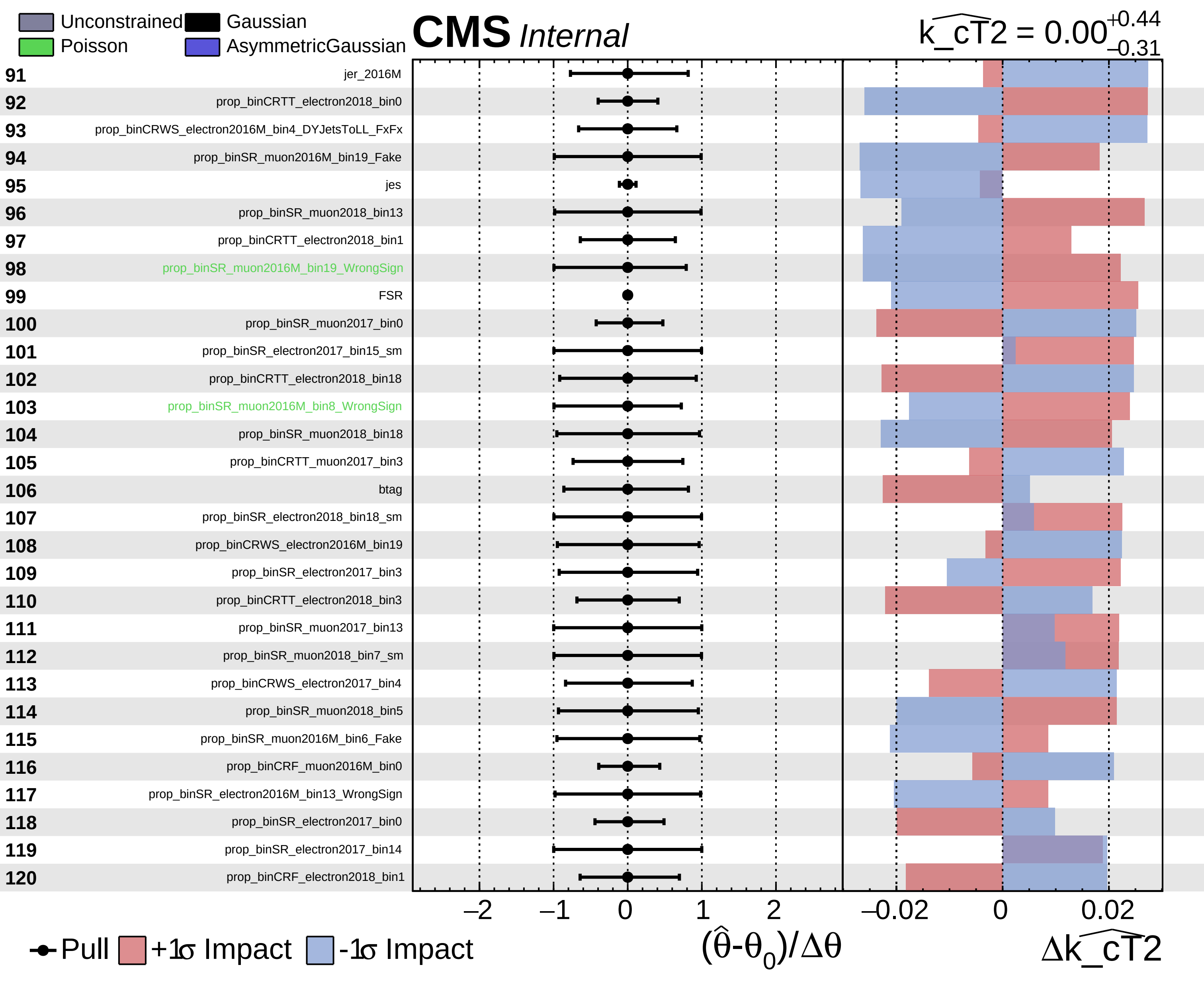
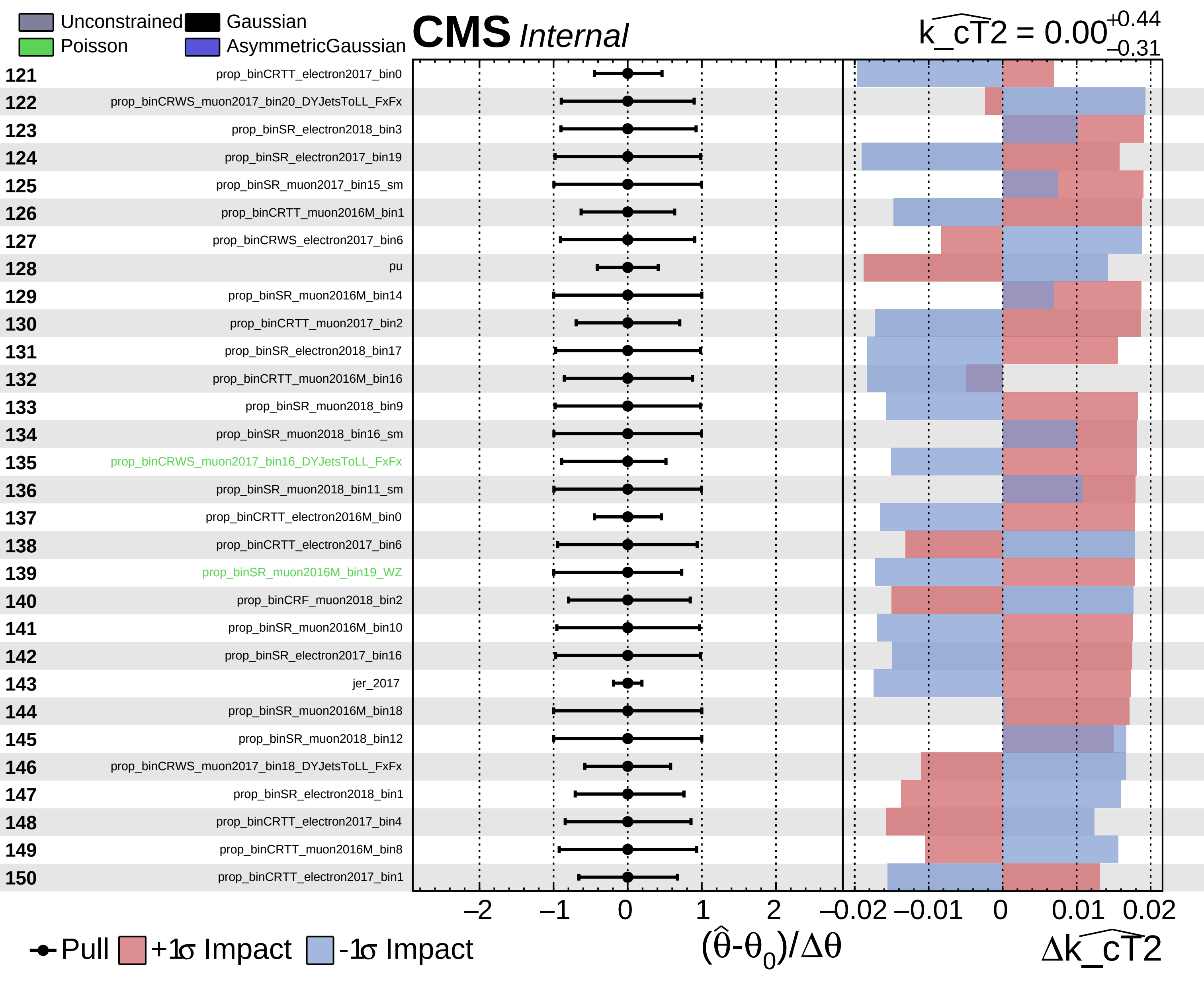

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
1 pdf_total
2 prop_binSR_electron2018_bin13
3 prop_binCRF_muon2018_bin0
4 prop_binSR_muon2018_bin20
5 prop_binCRF_muon2017_bin0
6 prop_binSR_electron2018_bin20
7 FR_sys_muon_2017
8 PF
9 prop_binSR_muon2018_bin19
10 prop_binSR_muon2017_bin19
11 prop_binCRWS_electron2017_bin2
12 prop_binSR_muon2017_bin16_sm
13 prop_binCRWS_muon2018_bin1
14 TTest_electron_2016M 1.00 +0.08 −0.08
15 prop_binSR_muon2017_bin20
16 prop_binCRF_muon2018_bin15
17 prop_binCRTT_muon2018_bin0
18 prop_binCRTT_muon2018_bin2
19 TTest_electron_2017 1.00 +0.06 −0.06
20 TTest_muon_2016M 1.00 +0.07 −0.07
21 TTest_muon_2017 1.00 +0.06 −0.06
22 DYest_electron_2016M 1.0 +0.5 −0.5
23 TTest_electron_2018 1.00 +0.06 −0.06
24 prop_binCRWS_muon2016M_bin1
25 TTest_muon_2018 1.00 +0.05 −0.05
26 prop_binCRWS_electron2018_bin3
27 prop_binCRWS_muon2017_bin1
28 DYest_muon_2017 1.0 +0.4 −0.4
29 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
30 prop_binCRWS_muon2018_bin3
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
31 DYest_muon_2016M $1.0^{+0.5}_{-0.5}$
32 prop_binCRWS_electron2018_bin1
33 DYest_muon_2018 $1.00^{+0.29}_{-0.29}$
34 DYest_electron_2017 $1.0^{+0.5}_{-0.5}$
35 prop_binCRWS_muon2018_bin2
36 prop_binCRWS_electron2017_bin1
37 prop_binSR_electron2017_bin20
38 DYest_electron_2018 $1.0^{+0.5}_{-0.5}$
39 prop_binSR_muon2016M_bin20
40 lep_2018
41 TES
42 lep_2016M
43 prop_binSR_electron2016M_bin20_Fake
44 prop_binSR_electron2016M_bin20_sm
45 prop_binCRWS_electron2018_bin2
46 prop_binCRWS_electron2018_bin0
47 prop_binCRWS_muon2017_bin2
48 prop_binCRWS_muon2016M_bin2
49 mistag
50 prop_binCRWS_muon2018_bin4
51 prop_binSR_muon2017_bin4
52 tau_vsjet_2016M
53 prop_binCRTT_muon2018_bin1
54 prop_binCRF_electron2017_bin0
55 FR_sys_muon_2018
56 prop_binCRWS_electron2017_bin3
57 prop_binCRTT_muon2017_bin20
58 prop_binCRF_electron2018_bin0
59 prop_binSR_electron2016M_bin20_TTTo2L2Nu
60 prop_binSR_muon2018_bin0
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

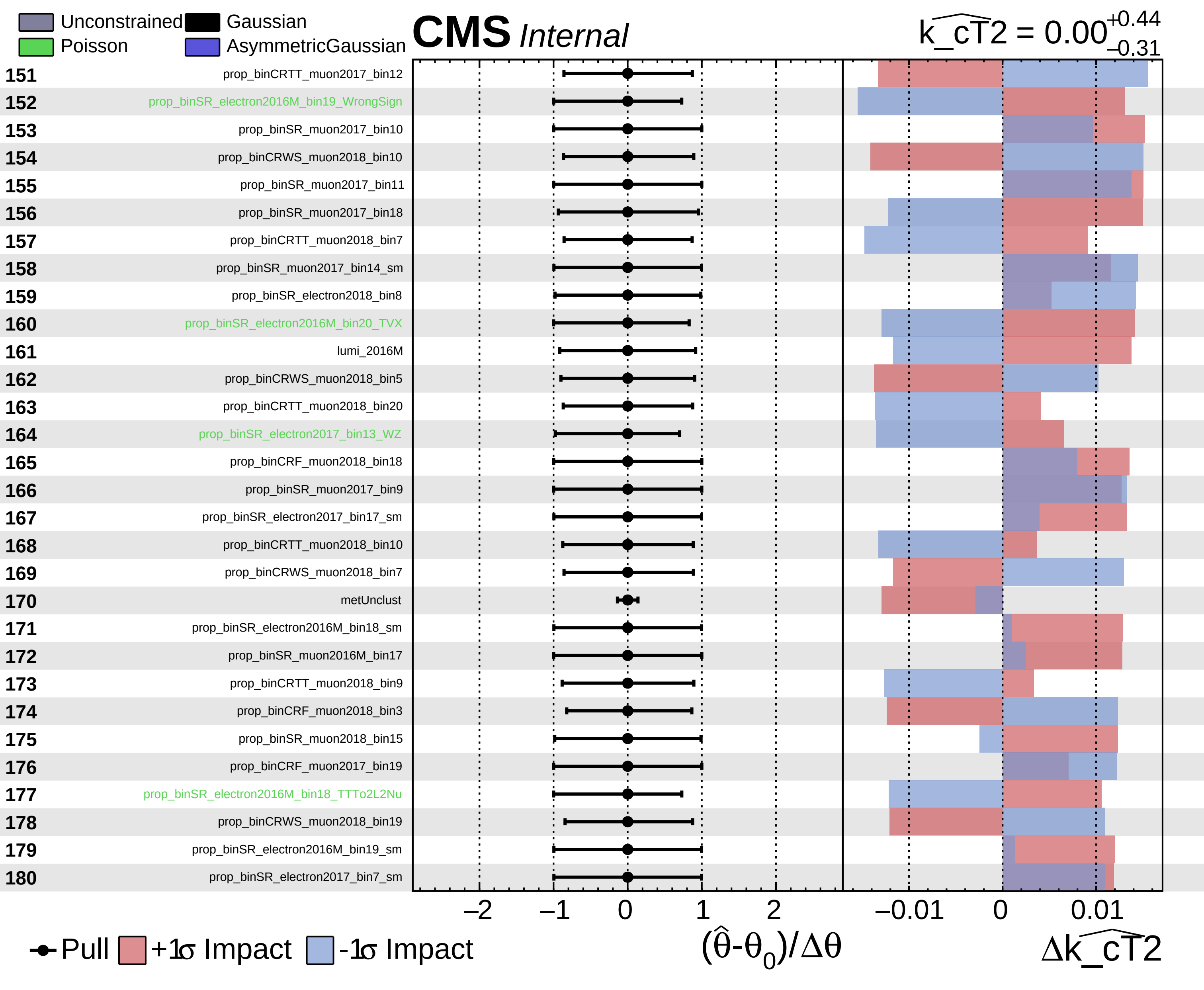

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
61 FR_sys_electron_2018
62 prop_binCRTT_electron2016M_bin16
63 prop_binCRWS_muon2016M_bin3
64 prop_binCRTT_muon2017_bin0
65 prop_binSR_muon2018_bin17
66 prop_binSR_muon2018_bin1
67 prop_binCRWS_electron2018_bin15_DYJetsToLL_FxFx
68 prop_binSR_electron2018_bin18_VG
69 prop_binCRWS_muon2018_bin16_DYJetsToLL_FxFx
70 prop_binCRWS_electron2016M_bin0
71 prop_binCRWS_muon2016M_bin4_DYJetsToLL_FxFx
72 prop_binCRF_muon2018_bin1
73 prop_binCRWS_electron2017_bin20_DYJetsToLL_FxFx
74 prop_binCRWS_muon2018_bin20_DYJetsToLL_FxFx
75 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
76 FR_sys_electron_2017
77 prop_binCRWS_muon2016M_bin13_DYJetsToLL_FxFx
78 prop_binCRWS_electron2017_bin0
79 prop_binCRWS_muon2016M_bin0
80 prop_binCRTT_muon2018_bin5
81 lep_2017
82 prop_binSR_electron2018_bin19
83 jer_2018
84 prop_binSR_electron2016M_bin20_ZZtoLep
85 prop_binSR_electron2018_bin16
86 prop_binSR_electron2018_bin0
87 ISR
88 prop_binCRTT_muon2018_bin3
89 prop_binSR_muon2018_bin14_sm
90 prop_binCRWS_electron2018_bin18_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

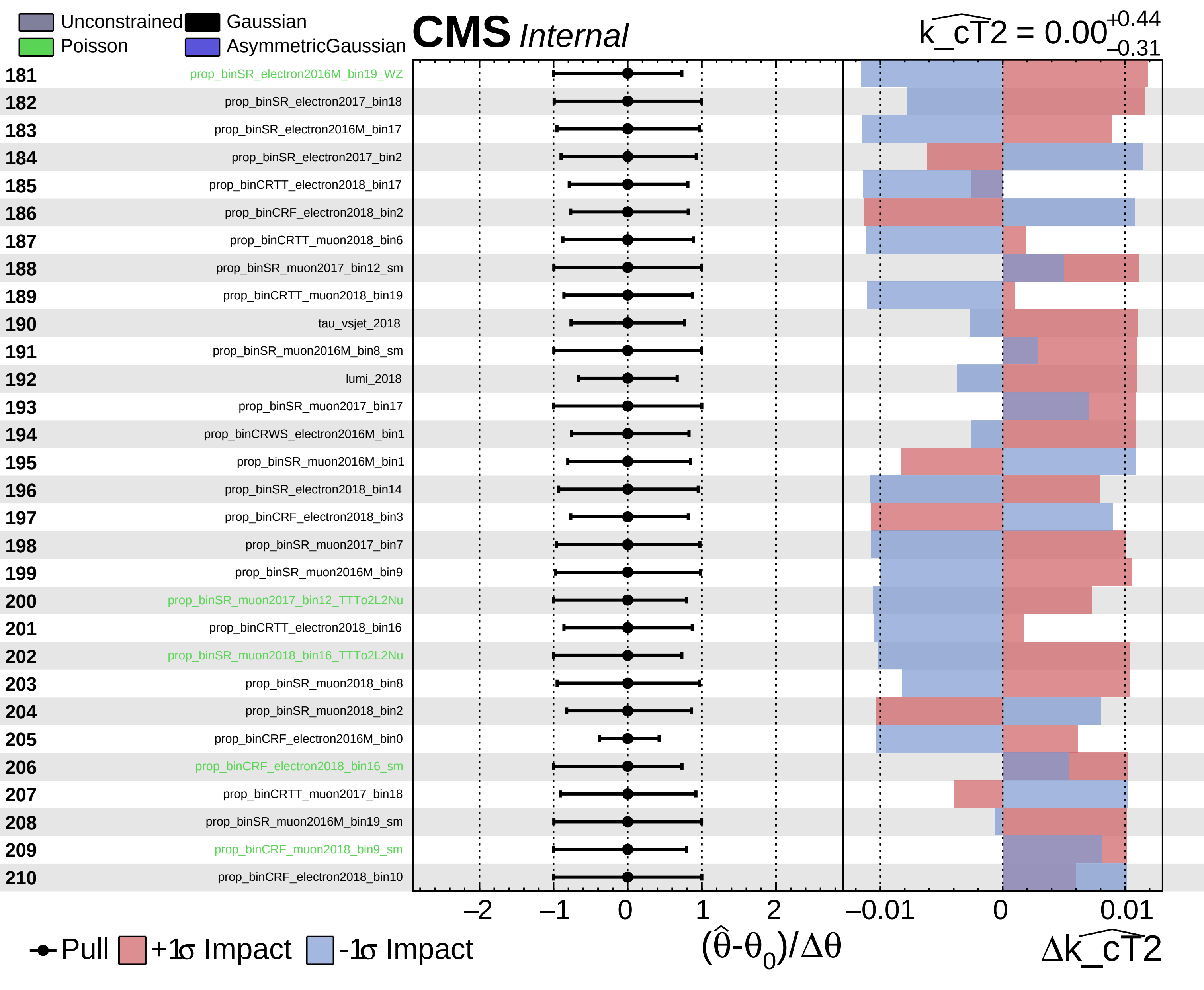


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
91 jer_2016M
92 prop_binCRTT_electron2018_bin0
93 prop_binCRWS_electron2016M_bin4_DYJetsToLL_FxFx
94 prop_binSR_muon2016M_bin19_Fake
95 jes
96 prop_binSR_muon2018_bin13
97 prop_binCRTT_electron2018_bin1
98 prop_binSR_muon2016M_bin19_WrongSign
99 FSR
100 prop_binSR_muon2017_bin0
101 prop_binSR_electron2017_bin15_sm
102 prop_binCRTT_electron2018_bin18
103 prop_binSR_muon2016M_bin8_WrongSign
104 prop_binSR_muon2018_bin18
105 prop_binCRTT_muon2017_bin3
106 btag
107 prop_binSR_electron2018_bin18_sm
108 prop_binCRWS_electron2016M_bin19
109 prop_binSR_electron2017_bin3
110 prop_binCRTT_electron2018_bin3
111 prop_binSR_muon2017_bin13
112 prop_binSR_muon2018_bin7_sm
113 prop_binCRWS_electron2017_bin4
114 prop_binSR_muon2018_bin5
115 prop_binSR_muon2016M_bin6_Fake
116 prop_binCRF_muon2016M_bin0
117 prop_binSR_electron2016M_bin13_WrongSign
118 prop_binSR_electron2017_bin0
119 prop_binSR_electron2017_bin14
120 prop_binCRF_electron2018_bin1
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

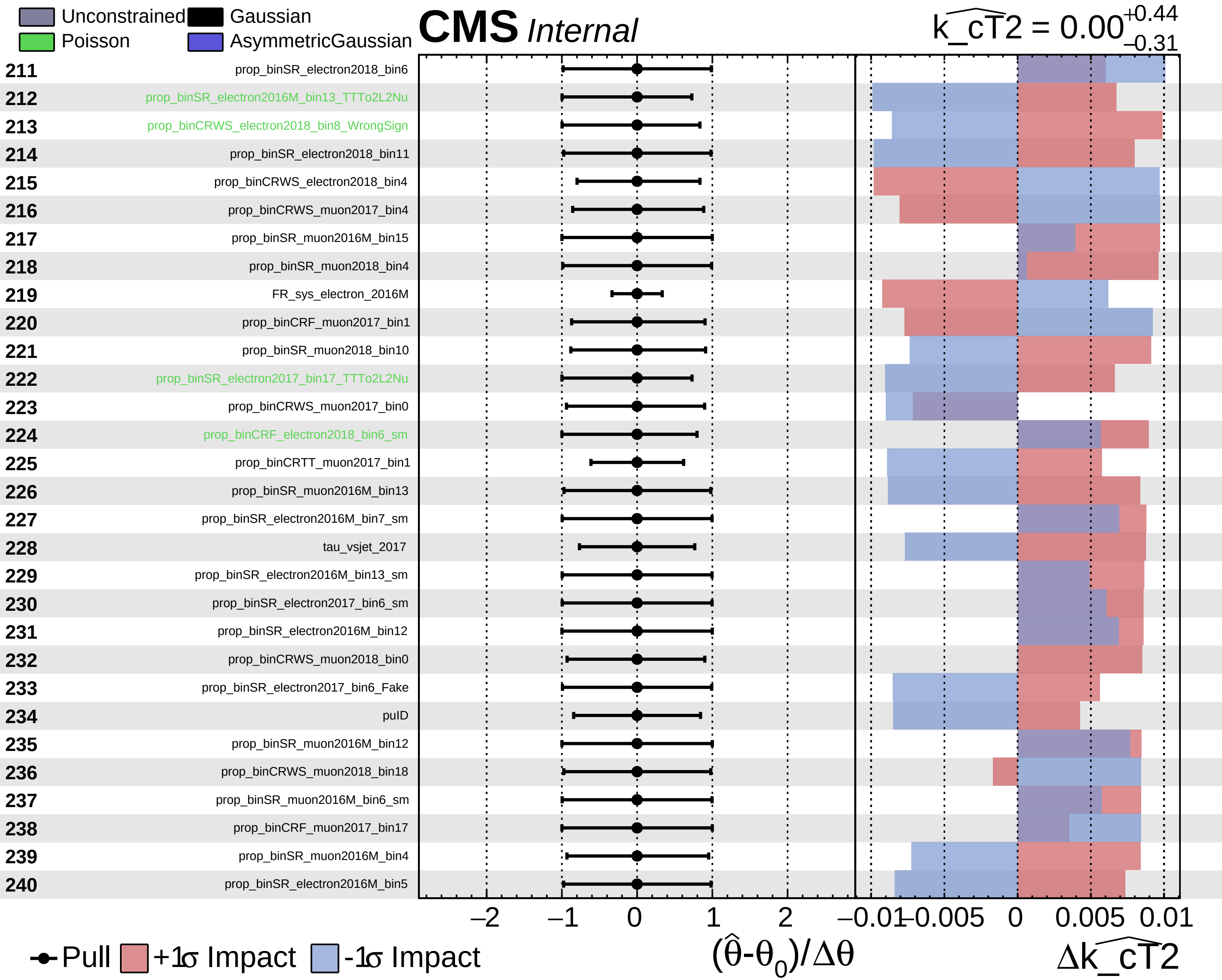


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = $0.00^{+0.44}_{-0.31}$
121 prop_binCRTT_electron2017_bin0
122 prop_binCRWS_muon2017_bin20_DYJetsToLL_FxFx
123 prop_binSR_electron2018_bin3
124 prop_binSR_electron2017_bin19
125 prop_binSR_muon2017_bin15_sm
126 prop_binCRTT_muon2016M_bin1
127 prop_binCRWS_electron2017_bin6
128 pu
129 prop_binSR_muon2016M_bin14
130 prop_binCRTT_muon2017_bin2
131 prop_binSR_electron2018_bin17
132 prop_binCRTT_muon2016M_bin16
133 prop_binSR_muon2018_bin9
134 prop_binSR_muon2018_bin16_sm
135 prop_binCRWS_muon2017_bin16_DYJetsToLL_FxFx
136 prop_binSR_muon2018_bin11_sm
137 prop_binCRTT_electron2016M_bin0
138 prop_binCRTT_electron2017_bin6
139 prop_binSR_muon2016M_bin19_WZ
140 prop_binCRF_muon2018_bin2
141 prop_binSR_muon2016M_bin10
142 prop_binSR_electron2017_bin16
143 jer_2017
144 prop_binSR_muon2016M_bin18
145 prop_binSR_muon2018_bin12
146 prop_binCRWS_muon2017_bin18_DYJetsToLL_FxFx
147 prop_binSR_electron2018_bin1
148 prop_binCRTT_electron2017_bin4
149 prop_binCRTT_muon2016M_bin8
150 prop_binCRTT_electron2017_bin1
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

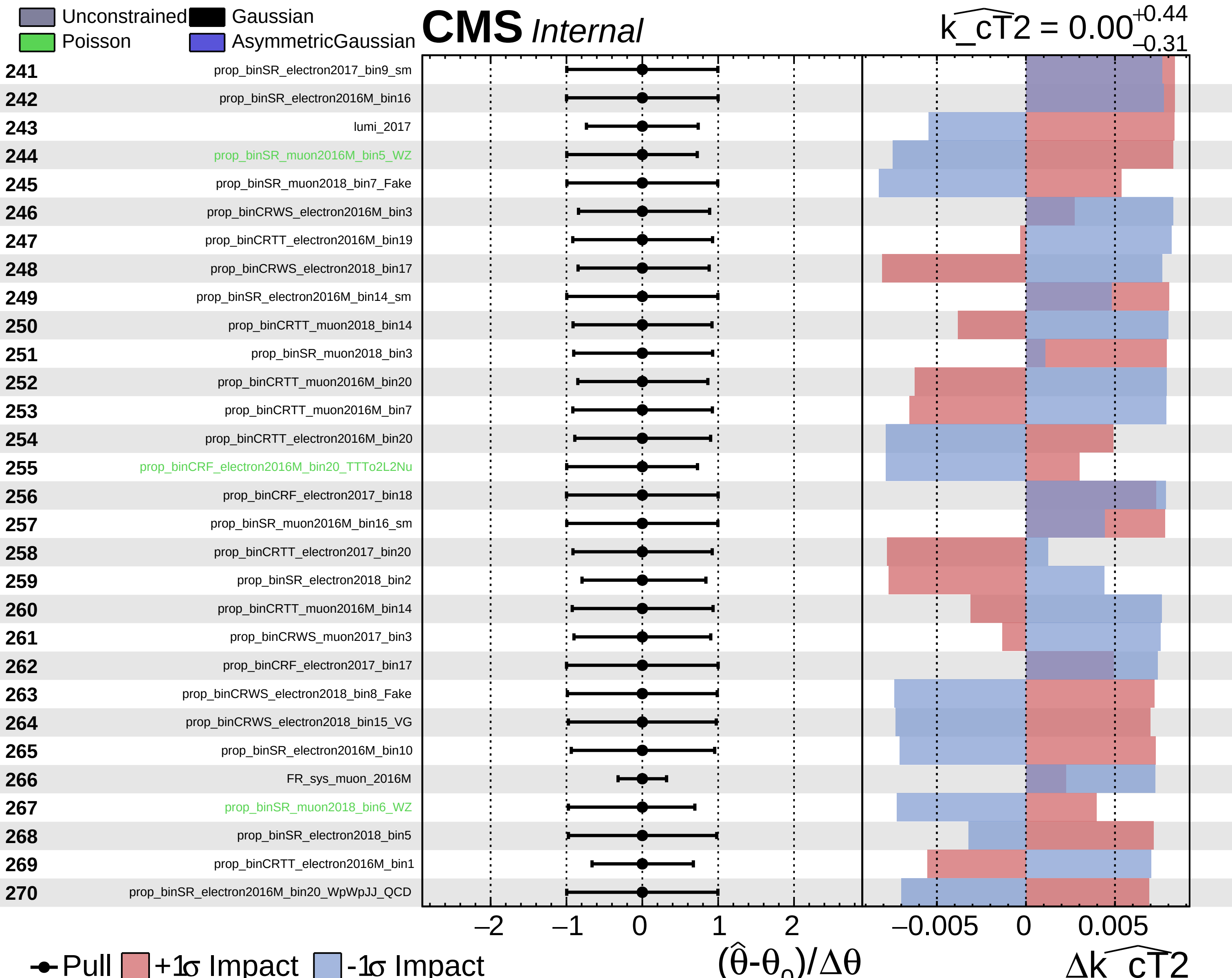

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
151 prop_binCRTT_muon2017_bin12
152 prop_binSR_electron2016M_bin19_WrongSign
153 prop_binSR_muon2017_bin10
154 prop_binCRWS_muon2018_bin10
155 prop_binSR_muon2017_bin11
156 prop_binSR_muon2017_bin18
157 prop_binCRTT_muon2018_bin7
158 prop_binSR_muon2017_bin14_sm
159 prop_binSR_electron2018_bin8
160 prop_binSR_electron2016M_bin20_TVX
161 lumi_2016M
162 prop_binCRWS_muon2018_bin5
163 prop_binCRTT_muon2018_bin20
164 prop_binSR_electron2017_bin13_WZ
165 prop_binCRF_muon2018_bin18
166 prop_binSR_muon2017_bin9
167 prop_binSR_electron2017_bin17_sm
168 prop_binCRTT_muon2018_bin10
169 prop_binCRWS_muon2018_bin7
170 metUnclust
171 prop_binSR_electron2016M_bin18_sm
172 prop_binSR_muon2016M_bin17
173 prop_binCRTT_muon2018_bin9
174 prop_binCRF_muon2018_bin3
175 prop_binSR_muon2018_bin15
176 prop_binCRF_muon2017_bin19
177 prop_binSR_electron2016M_bin18_TTTo2L2Nu
178 prop_binCRWS_muon2018_bin19
179 prop_binSR_electron2016M_bin19_sm
180 prop_binSR_electron2017_bin7_sm
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



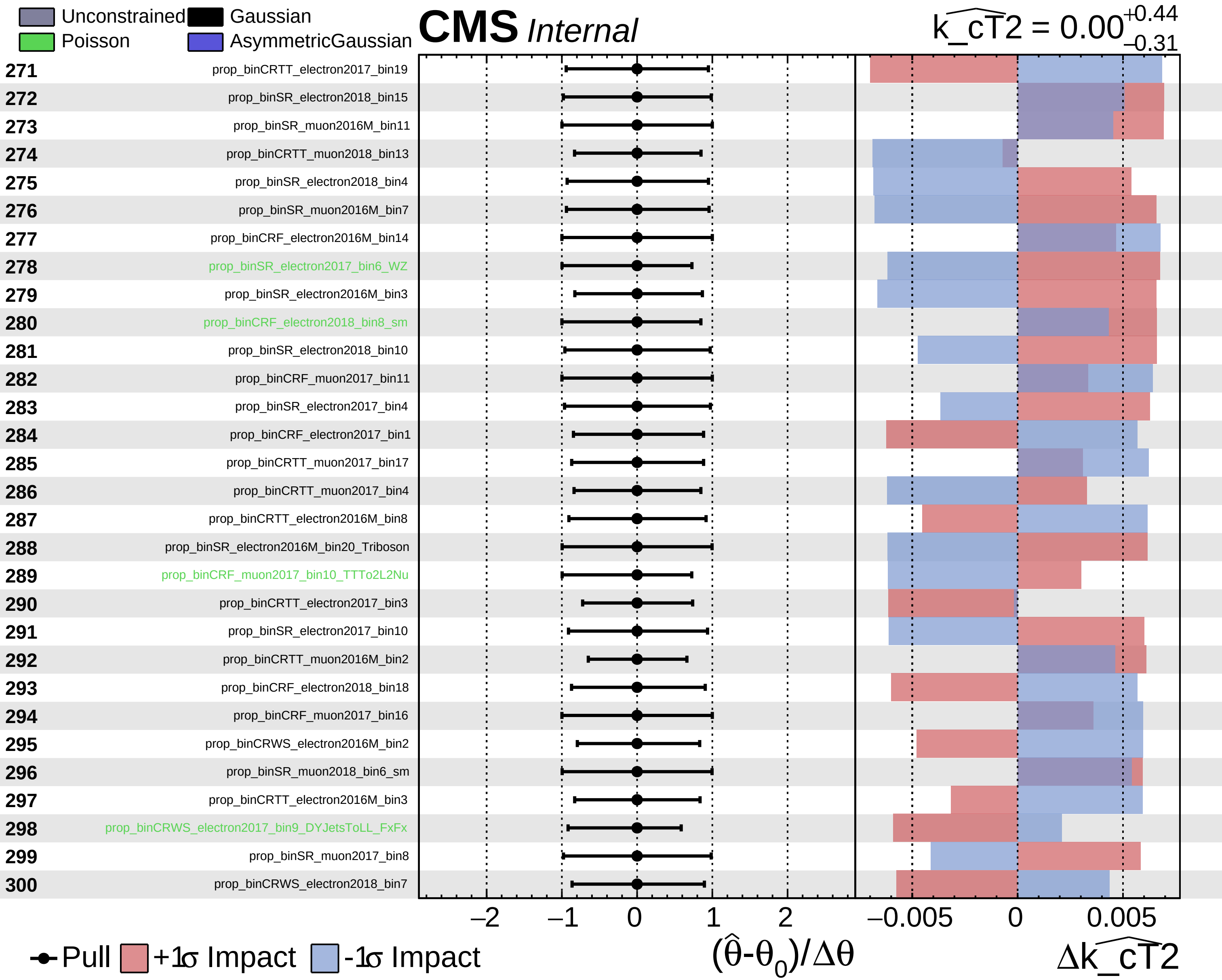
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
181 prop_binSR_electron2016M_bin19_WZ
182 prop_binSR_electron2017_bin18
183 prop_binSR_electron2016M_bin17
184 prop_binSR_electron2017_bin2
185 prop_binCRTT_electron2018_bin17
186 prop_binCRF_electron2018_bin2
187 prop_binCRTT_muon2018_bin6
188 prop_binSR_muon2017_bin12_sm
189 prop_binCRTT_muon2018_bin19
190 tau_vsjet_2018
191 prop_binSR_muon2016M_bin8_sm
192 lumi_2018
193 prop_binSR_muon2017_bin17
194 prop_binCRWS_electron2016M_bin1
195 prop_binSR_muon2016M_bin1
196 prop_binSR_electron2018_bin14
197 prop_binCRF_electron2018_bin3
198 prop_binSR_muon2017_bin7
199 prop_binSR_muon2016M_bin9
200 prop_binSR_muon2017_bin12_TTTo2L2Nu
201 prop_binCRTT_electron2018_bin16
202 prop_binSR_muon2018_bin16_TTTo2L2Nu
203 prop_binSR_muon2018_bin8
204 prop_binSR_muon2018_bin2
205 prop_binCRF_electron2016M_bin0
206 prop_binCRF_electron2018_bin16_sm
207 prop_binCRTT_muon2017_bin18
208 prop_binSR_muon2016M_bin19_sm
209 prop_binCRF_muon2018_bin9_sm
210 prop_binCRF_electron2018_bin10
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



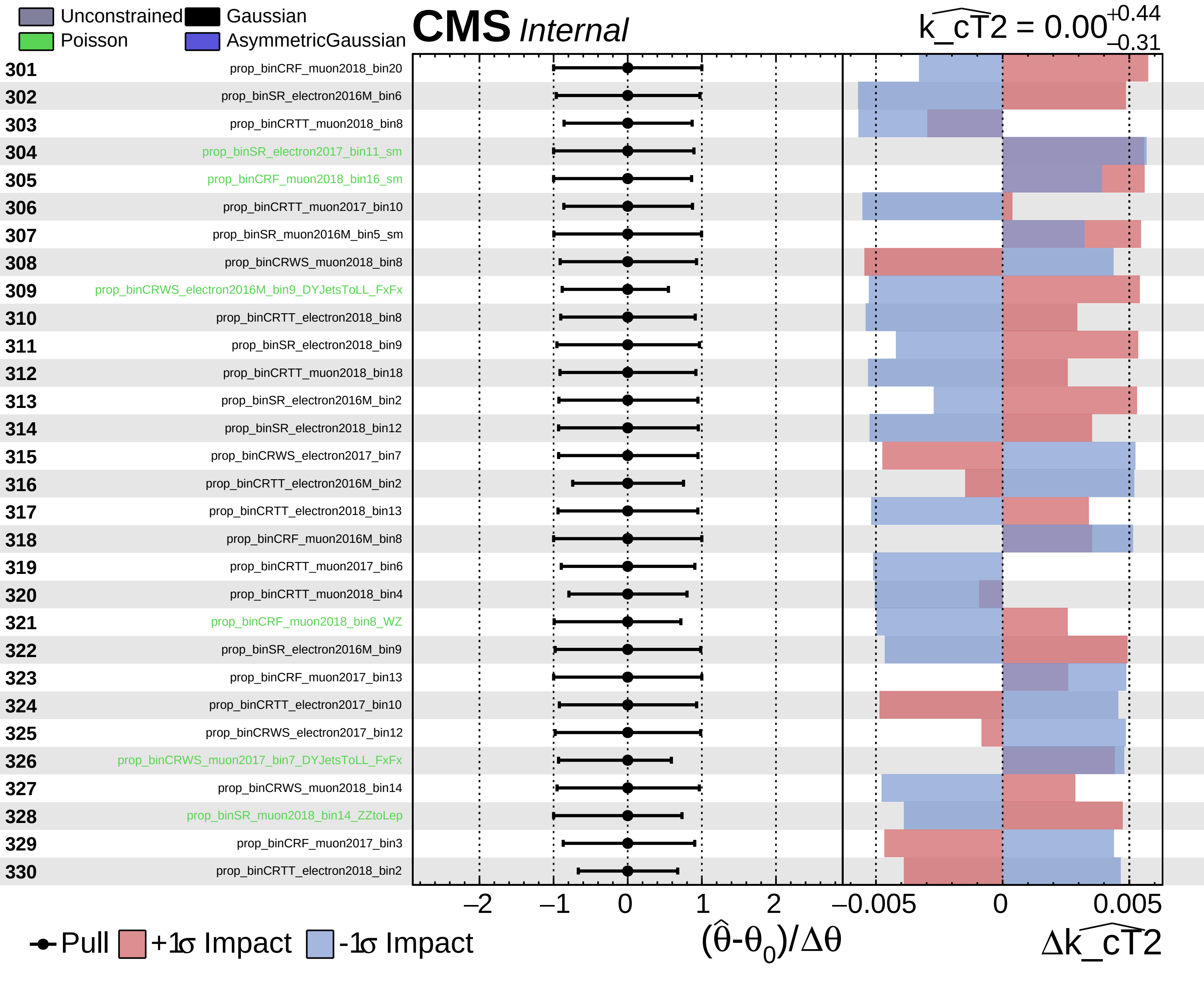
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
211 prop_binSR_electron2018_bin6
212 prop_binSR_electron2016M_bin13_TTTo2L2Nu
213 prop_binCRWS_electron2018_bin8_WrongSign
214 prop_binSR_electron2018_bin11
215 prop_binCRWS_electron2018_bin4
216 prop_binCRWS_muon2017_bin4
217 prop_binSR_muon2016M_bin15
218 prop_binSR_muon2018_bin4
219 FR_sys_electron_2016M
220 prop_binCRF_muon2017_bin1
221 prop_binSR_muon2018_bin10
222 prop_binSR_electron2017_bin17_TTTo2L2Nu
223 prop_binCRWS_muon2017_bin0
224 prop_binCRF_electron2018_bin6_sm
225 prop_binCRTT_muon2017_bin1
226 prop_binSR_muon2016M_bin13
227 prop_binSR_electron2016M_bin7_sm
228 tau_vsjet_2017
229 prop_binSR_electron2016M_bin13_sm
230 prop_binSR_electron2017_bin6_sm
231 prop_binSR_electron2016M_bin12
232 prop_binCRWS_muon2018_bin0
233 prop_binSR_electron2017_bin6_Fake
234 puID
235 prop_binSR_muon2016M_bin12
236 prop_binCRWS_muon2018_bin18
237 prop_binSR_muon2016M_bin6_sm
238 prop_binCRF_muon2017_bin17
239 prop_binSR_muon2016M_bin4
240 prop_binSR_electron2016M_bin5
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



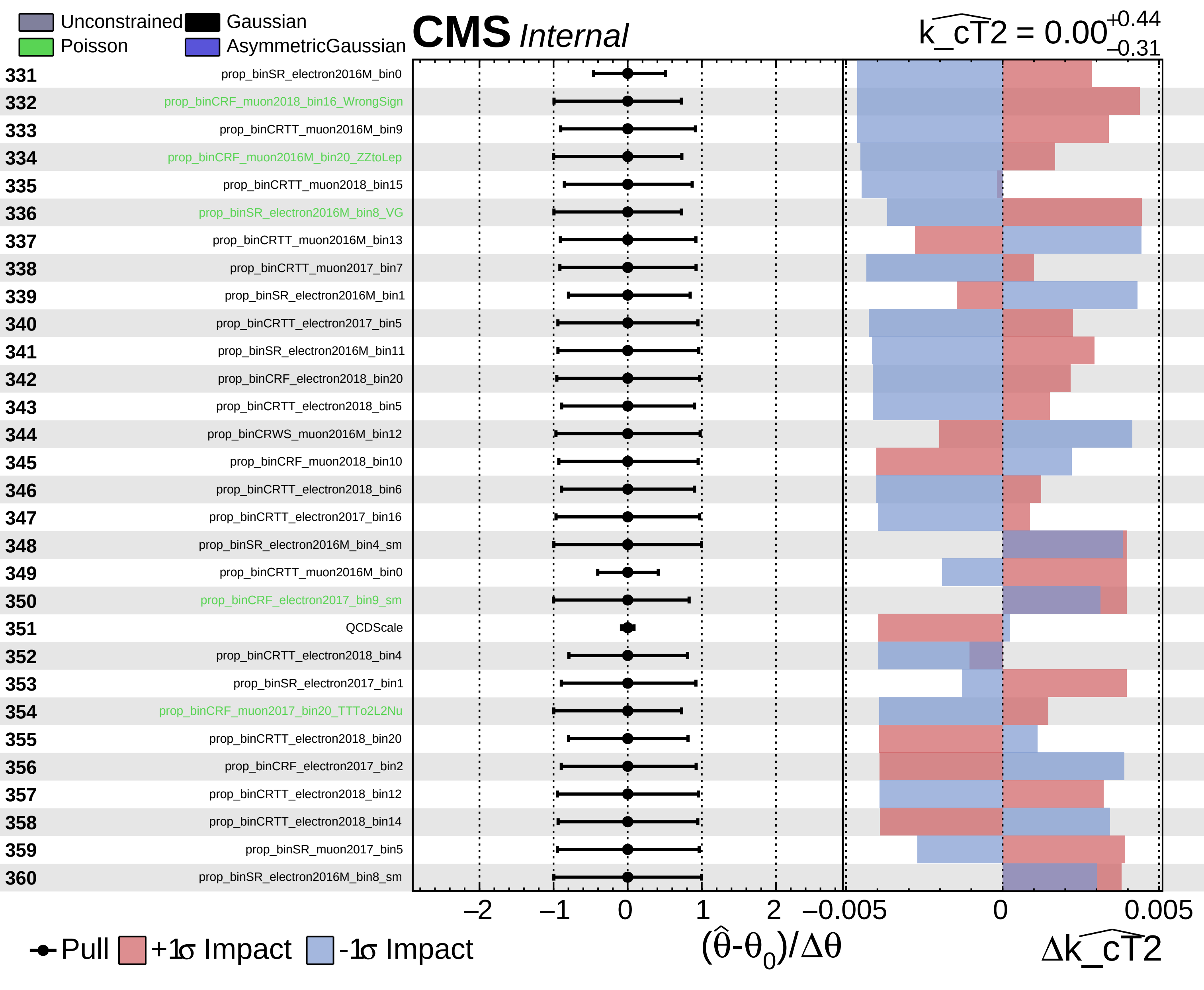
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
241 prop_binSR_electron2017_bin9_sm
242 prop_binSR_electron2016M_bin16
243 lumi_2017
244 prop_binSR_muon2016M_bin5_WZ
245 prop_binSR_muon2018_bin7_Fake
246 prop_binCRWS_electron2016M_bin3
247 prop_binCRTT_electron2016M_bin19
248 prop_binCRWS_electron2018_bin17
249 prop_binSR_electron2016M_bin14_sm
250 prop_binCRTT_muon2018_bin14
251 prop_binSR_muon2018_bin3
252 prop_binCRTT_muon2016M_bin20
253 prop_binCRTT_muon2016M_bin7
254 prop_binCRTT_electron2016M_bin20
255 prop_binCRF_electron2016M_bin20_TTTo2L2Nu
256 prop_binCRF_electron2017_bin18
257 prop_binSR_muon2016M_bin16_sm
258 prop_binCRTT_electron2017_bin20
259 prop_binSR_electron2018_bin2
260 prop_binCRTT_muon2016M_bin14
261 prop_binCRWS_muon2017_bin3
262 prop_binCRF_electron2017_bin17
263 prop_binCRWS_electron2018_bin8_Fake
264 prop_binCRWS_electron2018_bin15_VG
265 prop_binSR_electron2016M_bin10
266 FR_sys_muon_2016M
267 prop_binSR_muon2018_bin6_WZ
268 prop_binSR_electron2018_bin5
269 prop_binCRTT_electron2016M_bin1
270 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



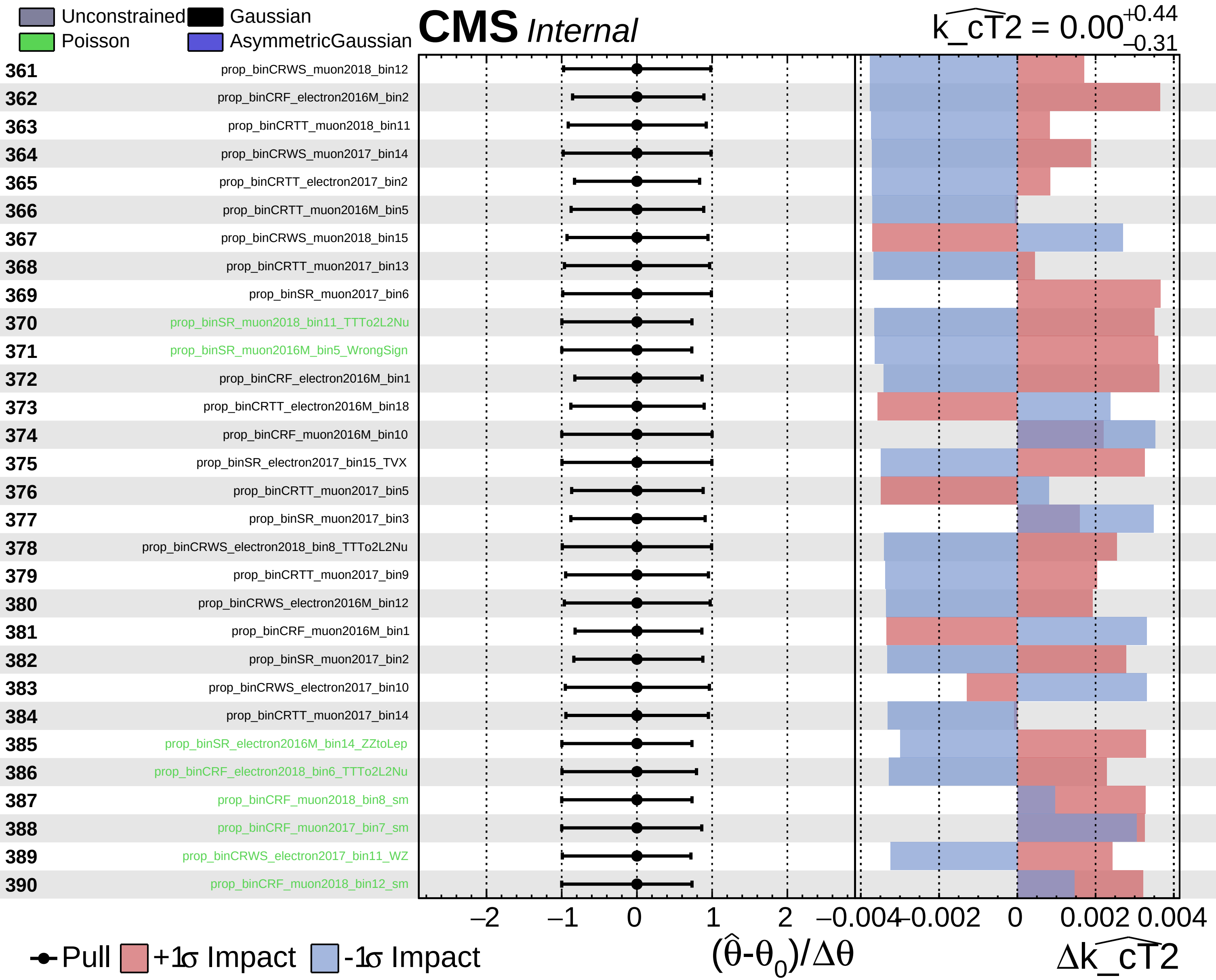
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
271 prop_binCRTT_electron2017_bin19
272 prop_binSR_electron2018_bin15
273 prop_binSR_muon2016M_bin11
274 prop_binCRTT_muon2018_bin13
275 prop_binSR_electron2018_bin4
276 prop_binSR_muon2016M_bin7
277 prop_binCRF_electron2016M_bin14
278 prop_binSR_electron2017_bin6_WZ
279 prop_binSR_electron2016M_bin3
280 prop_binCRF_electron2018_bin8_sm
281 prop_binSR_electron2018_bin10
282 prop_binCRF_muon2017_bin11
283 prop_binSR_electron2017_bin4
284 prop_binCRF_electron2017_bin1
285 prop_binCRTT_muon2017_bin17
286 prop_binCRTT_muon2017_bin4
287 prop_binCRTT_electron2016M_bin8
288 prop_binSR_electron2016M_bin20_Triboson
289 prop_binCRF_muon2017_bin10_TTTo2L2Nu
290 prop_binCRTT_electron2017_bin3
291 prop_binSR_electron2017_bin10
292 prop_binCRTT_muon2016M_bin2
293 prop_binCRF_electron2018_bin18
294 prop_binCRF_muon2017_bin16
295 prop_binCRWS_electron2016M_bin2
296 prop_binSR_muon2018_bin6_sm
297 prop_binCRTT_electron2016M_bin3
298 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
299 prop_binSR_muon2017_bin8
300 prop_binCRWS_electron2018_bin7
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



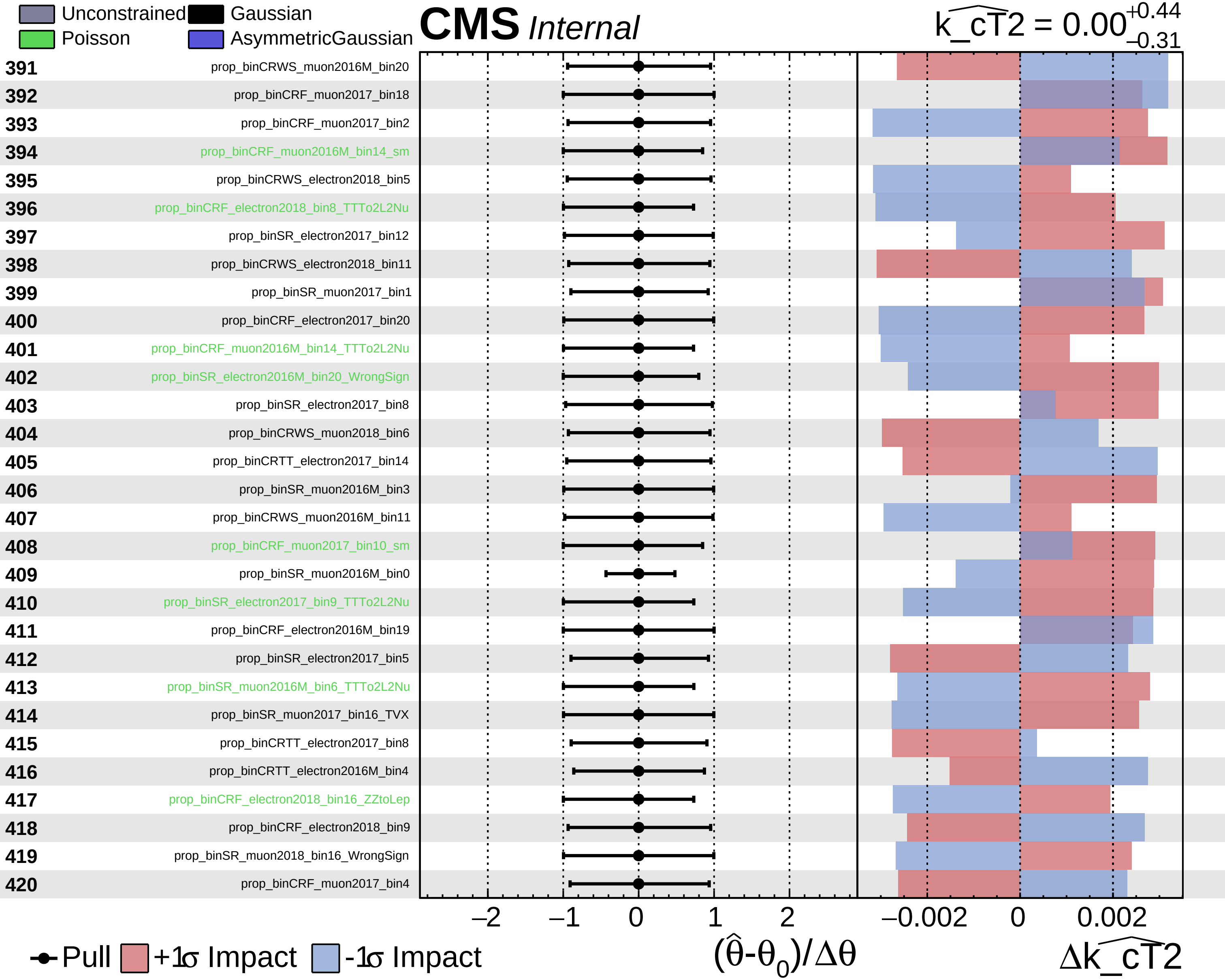
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
301 prop_binCRF_muon2018_bin20
302 prop_binSR_electron2016M_bin6
303 prop_binCRTT_muon2018_bin8
304 prop_binSR_electron2017_bin11_sm
305 prop_binCRF_muon2018_bin16_sm
306 prop_binCRTT_muon2017_bin10
307 prop_binSR_muon2016M_bin5_sm
308 prop_binCRWS_muon2018_bin8
309 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
310 prop_binCRTT_electron2018_bin8
311 prop_binSR_electron2018_bin9
312 prop_binCRTT_muon2018_bin18
313 prop_binSR_electron2016M_bin2
314 prop_binSR_electron2018_bin12
315 prop_binCRWS_electron2017_bin7
316 prop_binCRTT_electron2016M_bin2
317 prop_binCRTT_electron2018_bin13
318 prop_binCRF_muon2016M_bin8
319 prop_binCRTT_muon2017_bin6
320 prop_binCRTT_muon2018_bin4
321 prop_binCRF_muon2018_bin8_WZ
322 prop_binSR_electron2016M_bin9
323 prop_binCRF_muon2017_bin13
324 prop_binCRTT_electron2017_bin10
325 prop_binCRWS_electron2017_bin12
326 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
327 prop_binCRWS_muon2018_bin14
328 prop_binSR_muon2018_bin14_ZZtoLep
329 prop_binCRF_muon2017_bin3
330 prop_binCRTT_electron2018_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



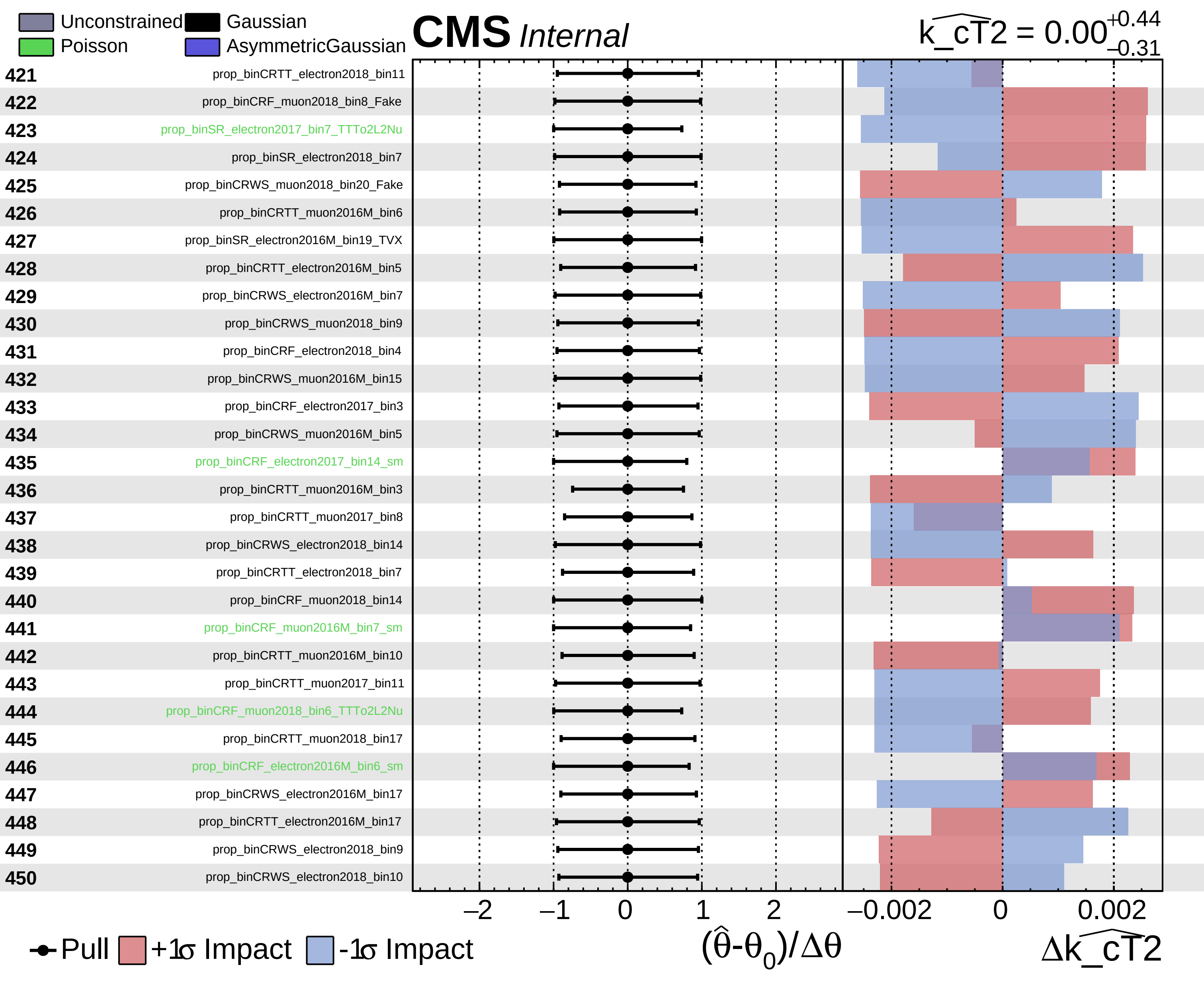
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
331 prop_binSR_electron2016M_bin0
332 prop_binCRF_muon2018_bin16_WrongSign
333 prop_binCRTT_muon2016M_bin9
334 prop_binCRF_muon2016M_bin20_ZZtoLep
335 prop_binCRTT_muon2018_bin15
336 prop_binSR_electron2016M_bin8_VG
337 prop_binCRTT_muon2016M_bin13
338 prop_binCRTT_muon2017_bin7
339 prop_binSR_electron2016M_bin1
340 prop_binCRTT_electron2017_bin5
341 prop_binSR_electron2016M_bin11
342 prop_binCRF_electron2018_bin20
343 prop_binCRTT_electron2018_bin5
344 prop_binCRWS_muon2016M_bin12
345 prop_binCRF_muon2018_bin10
346 prop_binCRTT_electron2018_bin6
347 prop_binCRTT_electron2017_bin16
348 prop_binSR_electron2016M_bin4_sm
349 prop_binCRTT_muon2016M_bin0
350 prop_binCRF_electron2017_bin9_sm
351 QCDScale
352 prop_binCRTT_electron2018_bin4
353 prop_binSR_electron2017_bin1
354 prop_binCRF_muon2017_bin20_TTTo2L2Nu
355 prop_binCRTT_electron2018_bin20
356 prop_binCRF_electron2017_bin2
357 prop_binCRTT_electron2018_bin12
358 prop_binCRTT_electron2018_bin14
359 prop_binSR_muon2017_bin5
360 prop_binSR_electron2016M_bin8_sm
−2 −1 0 1 2 −0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

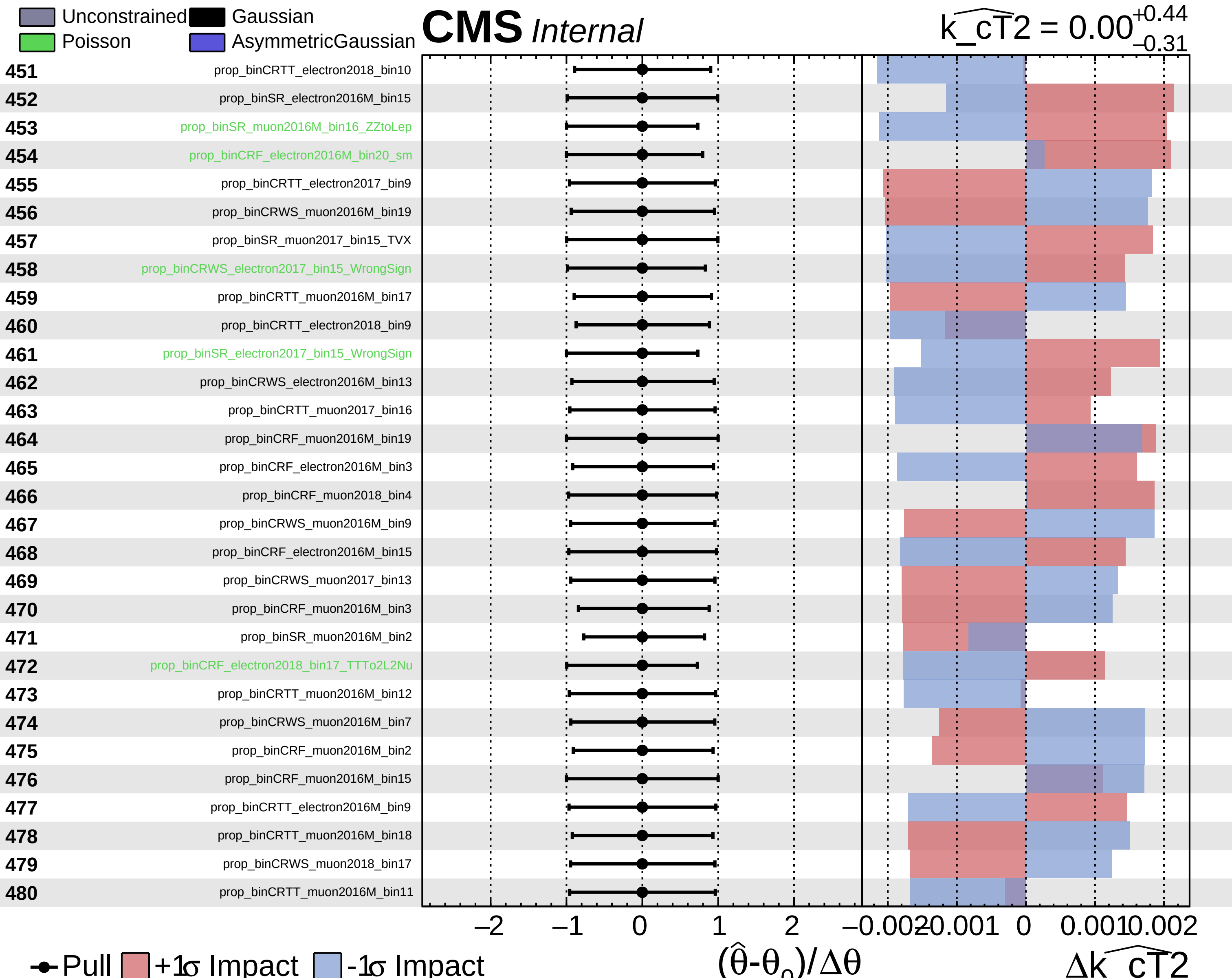


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
361 prop_binCRWS_muon2018_bin12
362 prop_binCRF_electron2016M_bin2
363 prop_binCRTT_muon2018_bin11
364 prop_binCRWS_muon2017_bin14
365 prop_binCRTT_electron2017_bin2
366 prop_binCRTT_muon2016M_bin5
367 prop_binCRWS_muon2018_bin15
368 prop_binCRTT_muon2017_bin13
369 prop_binSR_muon2017_bin6
370 prop_binSR_muon2018_bin11_TTTo2L2Nu
371 prop_binSR_muon2016M_bin5_WrongSign
372 prop_binCRF_electron2016M_bin1
373 prop_binCRTT_electron2016M_bin18
374 prop_binCRF_muon2016M_bin10
375 prop_binSR_electron2017_bin15_TVX
376 prop_binCRTT_muon2017_bin5
377 prop_binSR_muon2017_bin3
378 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
379 prop_binCRTT_muon2017_bin9
380 prop_binCRWS_electron2016M_bin12
381 prop_binCRF_muon2016M_bin1
382 prop_binSR_muon2017_bin2
383 prop_binCRWS_electron2017_bin10
384 prop_binCRTT_muon2017_bin14
385 prop_binSR_electron2016M_bin14_ZZtoLep
386 prop_binCRF_electron2018_bin6_TTTo2L2Nu
387 prop_binCRF_muon2018_bin8_sm
388 prop_binCRF_muon2017_bin7_sm
389 prop_binCRWS_electron2017_bin11_WZ
390 prop_binCRF_muon2018_bin12_sm
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

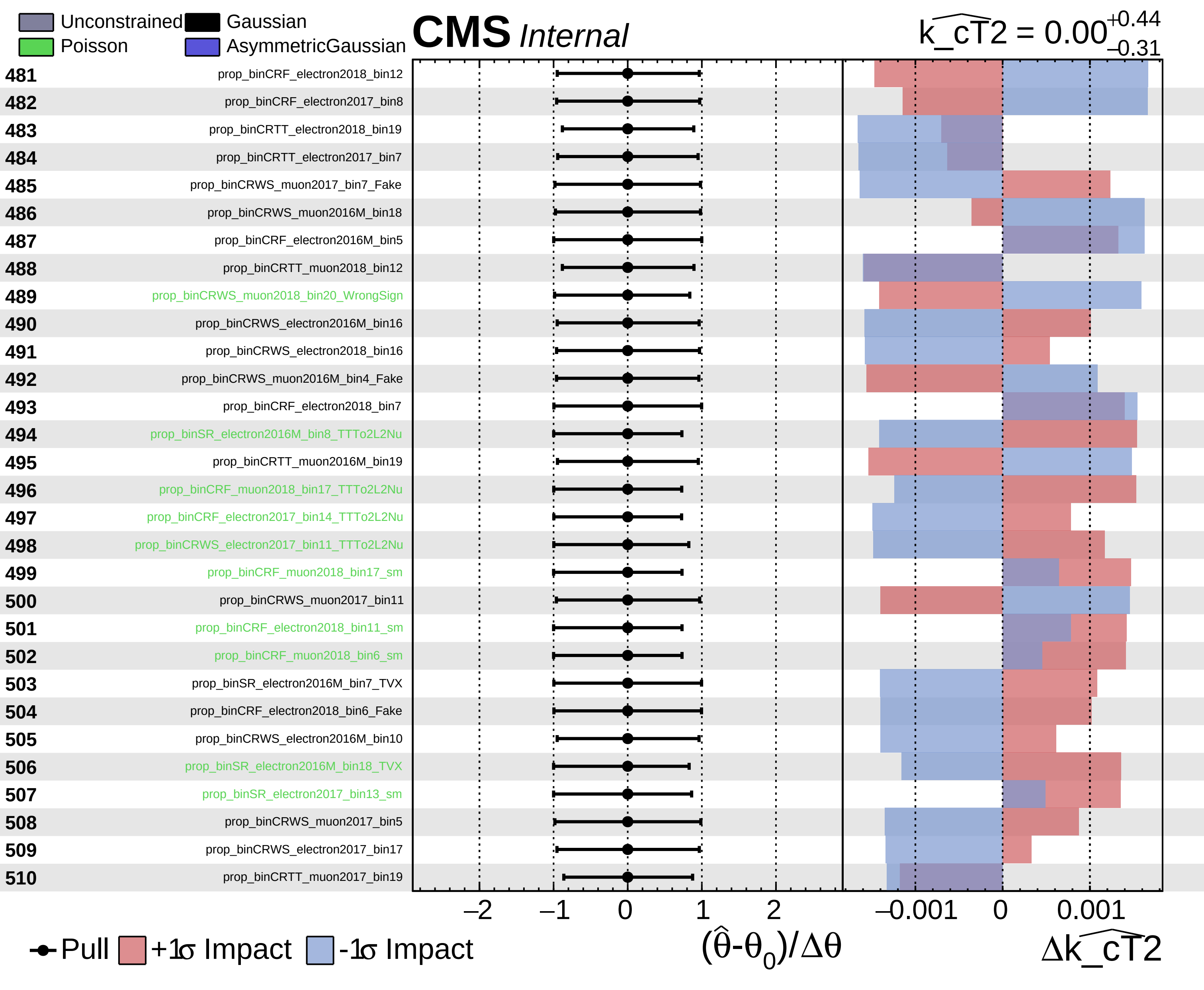


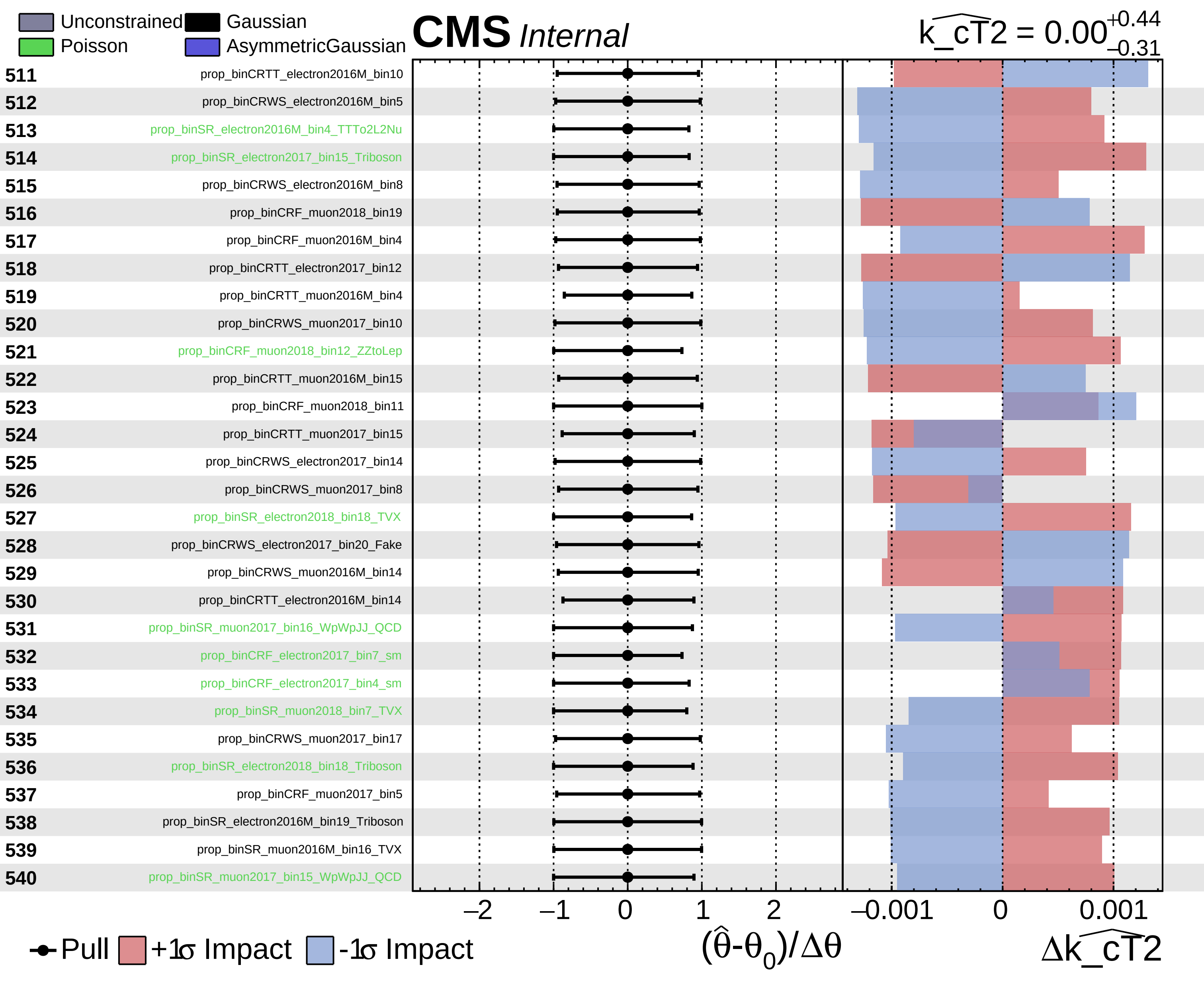
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
391 prop_binCRWS_muon2016M_bin20
392 prop_binCRF_muon2017_bin18
393 prop_binCRF_muon2017_bin2
394 prop_binCRF_muon2016M_bin14_sm
395 prop_binCRWS_electron2018_bin5
396 prop_binCRF_electron2018_bin8_TTTo2L2Nu
397 prop_binSR_electron2017_bin12
398 prop_binCRWS_electron2018_bin11
399 prop_binSR_muon2017_bin1
400 prop_binCRF_electron2017_bin20
401 prop_binCRF_muon2016M_bin14_TTTo2L2Nu
402 prop_binSR_electron2016M_bin20_WrongSign
403 prop_binSR_electron2017_bin8
404 prop_binCRWS_muon2018_bin6
405 prop_binCRTT_electron2017_bin14
406 prop_binSR_muon2016M_bin3
407 prop_binCRWS_muon2016M_bin11
408 prop_binCRF_muon2017_bin10_sm
409 prop_binSR_muon2016M_bin0
410 prop_binSR_electron2017_bin9_TTTo2L2Nu
411 prop_binCRF_electron2016M_bin19
412 prop_binSR_electron2017_bin5
413 prop_binSR_muon2016M_bin6_TTTo2L2Nu
414 prop_binSR_muon2017_bin16_TVX
415 prop_binCRTT_electron2017_bin8
416 prop_binCRTT_electron2016M_bin4
417 prop_binCRF_electron2018_bin16_ZZtoLep
418 prop_binCRF_electron2018_bin9
419 prop_binSR_muon2018_bin16_WrongSign
420 prop_binCRF_muon2017_bin4
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
421 prop_binCRTT_electron2018_bin11
422 prop_binCRF_muon2018_bin8_Fake
423 prop_binSR_electron2017_bin7_TTTo2L2Nu
424 prop_binSR_electron2018_bin7
425 prop_binCRWS_muon2018_bin20_Fake
426 prop_binCRTT_muon2016M_bin6
427 prop_binSR_electron2016M_bin19_TVX
428 prop_binCRTT_electron2016M_bin5
429 prop_binCRWS_electron2016M_bin7
430 prop_binCRWS_muon2018_bin9
431 prop_binCRF_electron2018_bin4
432 prop_binCRWS_muon2016M_bin15
433 prop_binCRF_electron2017_bin3
434 prop_binCRWS_muon2016M_bin5
435 prop_binCRF_electron2017_bin14_sm
436 prop_binCRTT_muon2016M_bin3
437 prop_binCRTT_muon2017_bin8
438 prop_binCRWS_electron2018_bin14
439 prop_binCRTT_electron2018_bin7
440 prop_binCRF_muon2018_bin14
441 prop_binCRF_muon2016M_bin7_sm
442 prop_binCRTT_muon2016M_bin10
443 prop_binCRTT_muon2017_bin11
444 prop_binCRF_muon2018_bin6_TTTo2L2Nu
445 prop_binCRTT_muon2018_bin17
446 prop_binCRF_electron2016M_bin6_sm
447 prop_binCRWS_electron2016M_bin17
448 prop_binCRTT_electron2016M_bin17
449 prop_binCRWS_electron2018_bin9
450 prop_binCRWS_electron2018_bin10
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

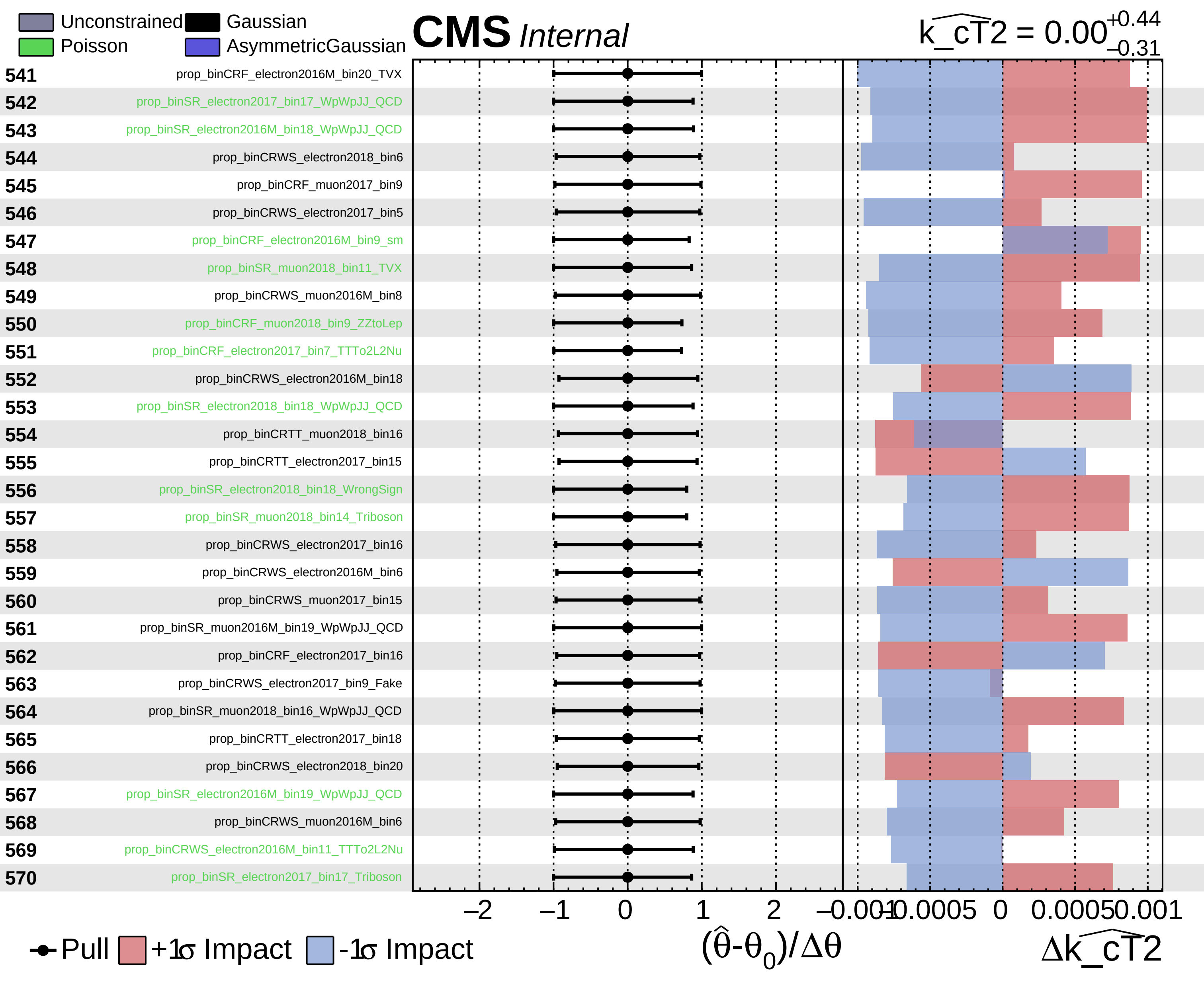


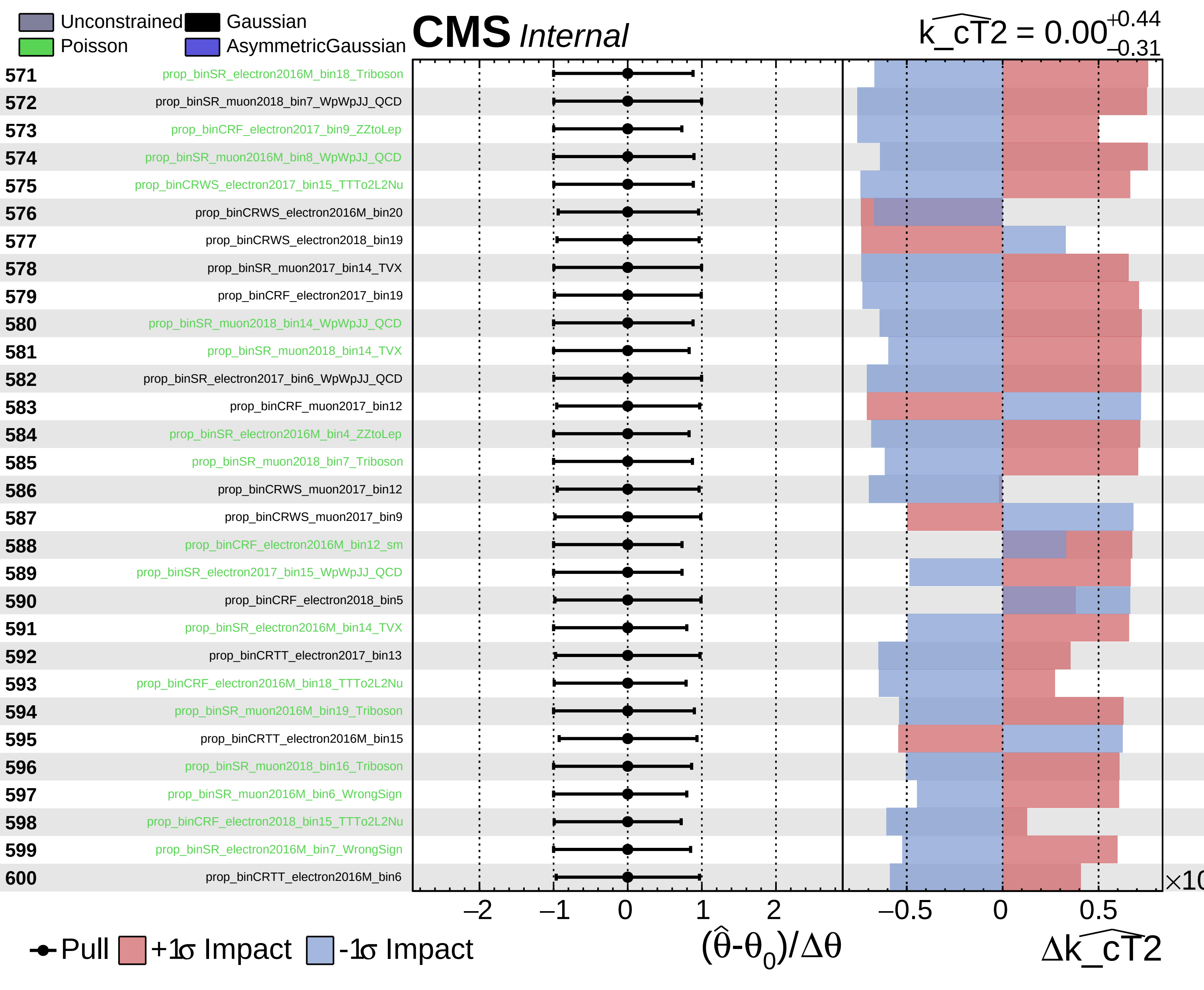
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
451 prop_binCRTT_electron2018_bin10
452 prop_binSR_electron2016M_bin15
453 prop_binSR_muon2016M_bin16_ZZtoLep
454 prop_binCRF_electron2016M_bin20_sm
455 prop_binCRTT_electron2017_bin9
456 prop_binCRWS_muon2016M_bin19
457 prop_binSR_muon2017_bin15_TVX
458 prop_binCRWS_electron2017_bin15_WrongSign
459 prop_binCRTT_muon2016M_bin17
460 prop_binCRTT_electron2018_bin9
461 prop_binSR_electron2017_bin15_WrongSign
462 prop_binCRWS_electron2016M_bin13
463 prop_binCRTT_muon2017_bin16
464 prop_binCRF_muon2016M_bin19
465 prop_binCRF_electron2016M_bin3
466 prop_binCRF_muon2018_bin4
467 prop_binCRWS_muon2016M_bin9
468 prop_binCRF_electron2016M_bin15
469 prop_binCRWS_muon2017_bin13
470 prop_binCRF_muon2016M_bin3
471 prop_binSR_muon2016M_bin2
472 prop_binCRF_electron2018_bin17_TTTo2L2Nu
473 prop_binCRTT_muon2016M_bin12
474 prop_binCRWS_muon2016M_bin7
475 prop_binCRF_muon2016M_bin2
476 prop_binCRF_muon2016M_bin15
477 prop_binCRTT_electron2016M_bin9
478 prop_binCRTT_muon2016M_bin18
479 prop_binCRWS_muon2018_bin17
480 prop_binCRTT_muon2016M_bin11
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
481 prop_binCRF_electron2018_bin12
482 prop_binCRF_electron2017_bin8
483 prop_binCRTT_electron2018_bin19
484 prop_binCRTT_electron2017_bin7
485 prop_binCRWS_muon2017_bin7_Fake
486 prop_binCRWS_muon2016M_bin18
487 prop_binCRF_electron2016M_bin5
488 prop_binCRTT_muon2018_bin12
489 prop_binCRWS_muon2018_bin20_WrongSign
490 prop_binCRWS_electron2016M_bin16
491 prop_binCRWS_electron2018_bin16
492 prop_binCRWS_muon2016M_bin4_Fake
493 prop_binCRF_electron2018_bin7
494 prop_binSR_electron2016M_bin8_TTTo2L2Nu
495 prop_binCRTT_muon2016M_bin19
496 prop_binCRF_muon2018_bin17_TTTo2L2Nu
497 prop_binCRF_electron2017_bin14_TTTo2L2Nu
498 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
499 prop_binCRF_muon2018_bin17_sm
500 prop_binCRWS_muon2017_bin11
501 prop_binCRF_electron2018_bin11_sm
502 prop_binCRF_muon2018_bin6_sm
503 prop_binSR_electron2016M_bin7_TVX
504 prop_binCRF_electron2018_bin6_Fake
505 prop_binCRWS_electron2016M_bin10
506 prop_binSR_electron2016M_bin18_TVX
507 prop_binSR_electron2017_bin13_sm
508 prop_binCRWS_muon2017_bin5
509 prop_binCRWS_electron2017_bin17
510 prop_binCRTT_muon2017_bin19
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

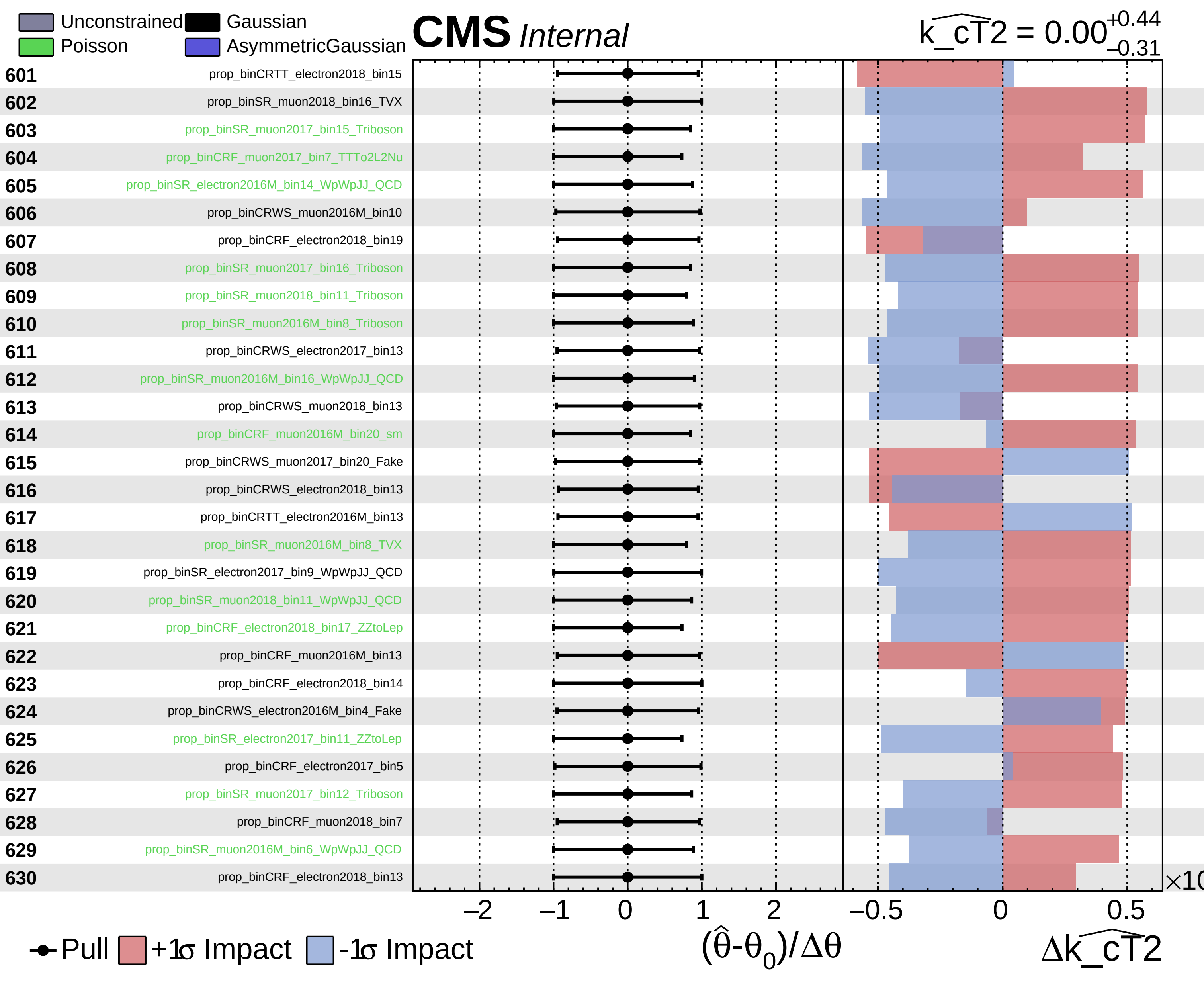


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
511 prop_binCRTT_electron2016M_bin10
512 prop_binCRWS_electron2016M_bin5
513 prop_binSR_electron2016M_bin4_TTTo2L2Nu
514 prop_binSR_electron2017_bin15_Triboson
515 prop_binCRWS_electron2016M_bin8
516 prop_binCRF_muon2018_bin19
517 prop_binCRF_muon2016M_bin4
518 prop_binCRTT_electron2017_bin12
519 prop_binCRTT_muon2016M_bin4
520 prop_binCRWS_muon2017_bin10
521 prop_binCRF_muon2018_bin12_ZZtoLep
522 prop_binCRTT_muon2016M_bin15
523 prop_binCRF_muon2018_bin11
524 prop_binCRTT_muon2017_bin15
525 prop_binCRWS_electron2017_bin14
526 prop_binCRWS_muon2017_bin8
527 prop_binSR_electron2018_bin18_TVX
528 prop_binCRWS_electron2017_bin20_Fake
529 prop_binCRWS_muon2016M_bin14
530 prop_binCRTT_electron2016M_bin14
531 prop_binSR_muon2017_bin16_WpWpJJ_QCD
532 prop_binCRF_electron2017_bin7_sm
533 prop_binCRF_electron2017_bin4_sm
534 prop_binSR_muon2018_bin7_TVX
535 prop_binCRWS_muon2017_bin17
536 prop_binSR_electron2018_bin18_Triboson
537 prop_binCRF_muon2017_bin5
538 prop_binSR_electron2016M_bin19_Triboson
539 prop_binSR_muon2016M_bin16_TVX
540 prop_binSR_muon2017_bin15_WpWpJJ_QCD
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

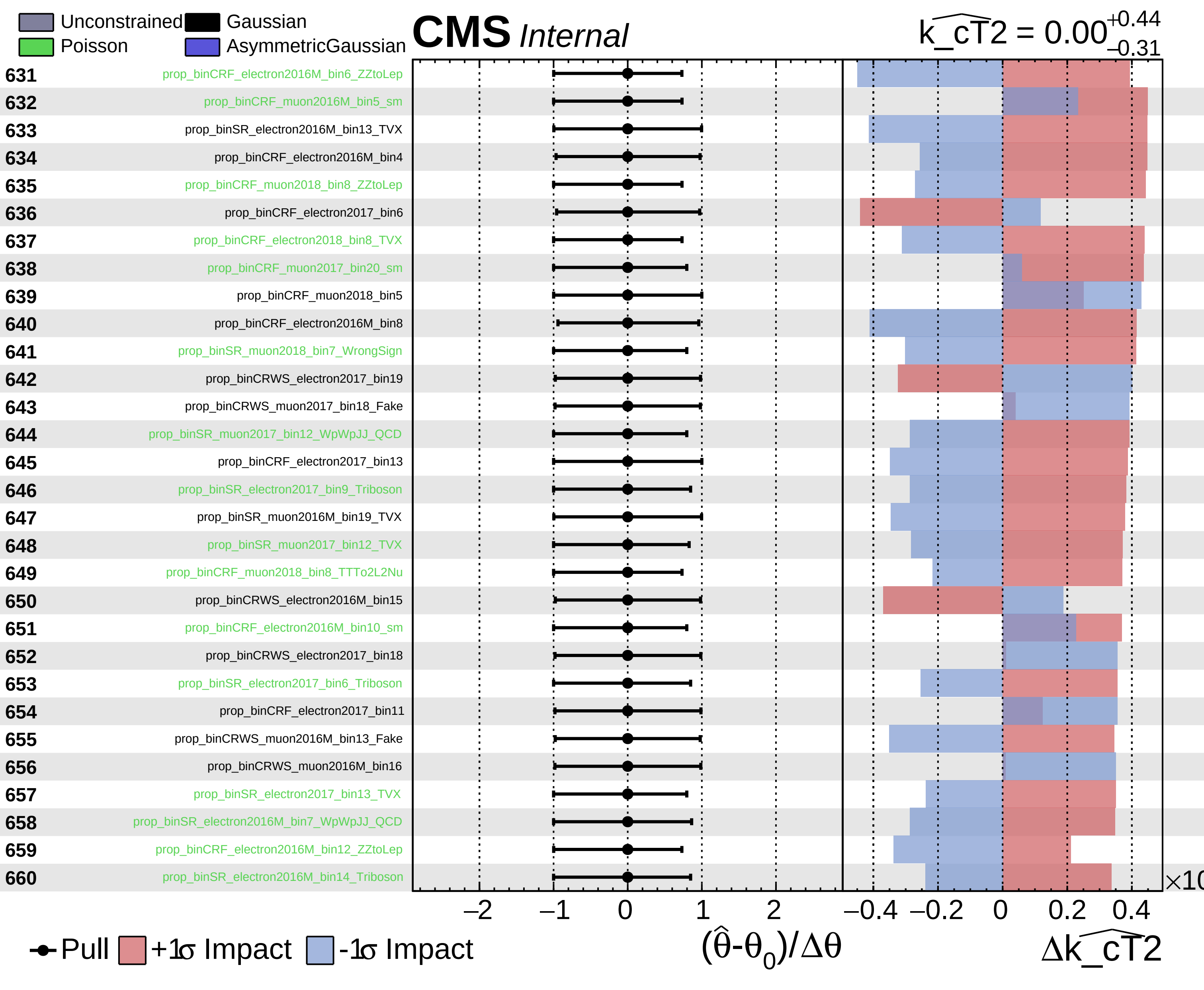

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
541 prop_binCRF_electron2016M_bin20_TVX
542 prop_binSR_electron2017_bin17_WpWpJJ_QCD
543 prop_binSR_electron2016M_bin18_WpWpJJ_QCD
544 prop_binCRWS_electron2018_bin6
545 prop_binCRF_muon2017_bin9
546 prop_binCRWS_electron2017_bin5
547 prop_binCRF_electron2016M_bin9_sm
548 prop_binSR_muon2018_bin11_TVX
549 prop_binCRWS_muon2016M_bin8
550 prop_binCRF_muon2018_bin9_ZZtoLep
551 prop_binCRF_electron2017_bin7_TTTo2L2Nu
552 prop_binCRWS_electron2016M_bin18
553 prop_binSR_electron2018_bin18_WpWpJJ_QCD
554 prop_binCRTT_muon2018_bin16
555 prop_binCRTT_electron2017_bin15
556 prop_binSR_electron2018_bin18_WrongSign
557 prop_binSR_muon2018_bin14_Triboson
558 prop_binCRWS_electron2017_bin16
559 prop_binCRWS_electron2016M_bin6
560 prop_binCRWS_muon2017_bin15
561 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
562 prop_binCRF_electron2017_bin16
563 prop_binCRWS_electron2017_bin9_Fake
564 prop_binSR_muon2018_bin16_WpWpJJ_QCD
565 prop_binCRTT_electron2017_bin18
566 prop_binCRWS_electron2018_bin20
567 prop_binSR_electron2016M_bin19_WpWpJJ_QCD
568 prop_binCRWS_muon2016M_bin6
569 prop_binCRWS_electron2016M_bin11_TTTo2L2Nu
570 prop_binSR_electron2017_bin17_Triboson
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



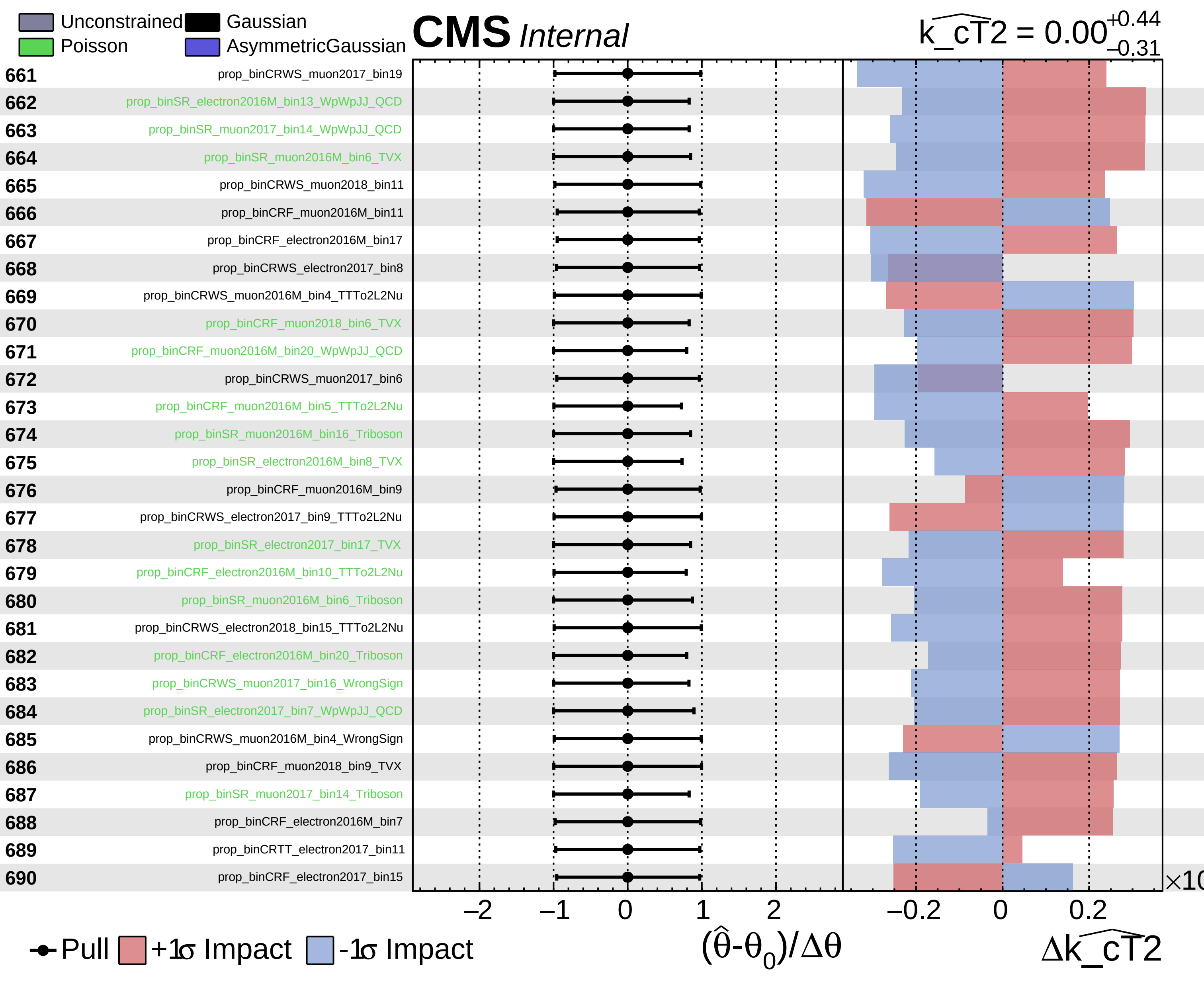
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
571 prop_binSR_electron2016M_bin18_Triboson
572 prop_binSR_muon2018_bin7_WpWpJJ_QCD
573 prop_binCRF_electron2017_bin9_ZZtoLep
574 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
575 prop_binCRWS_electron2017_bin15_TTTo2L2Nu
576 prop_binCRWS_electron2016M_bin20
577 prop_binCRWS_electron2018_bin19
578 prop_binSR_muon2017_bin14_TVX
579 prop_binCRF_electron2017_bin19
580 prop_binSR_muon2018_bin14_WpWpJJ_QCD
581 prop_binSR_muon2018_bin14_TVX
582 prop_binSR_electron2017_bin6_WpWpJJ_QCD
583 prop_binCRF_muon2017_bin12
584 prop_binSR_electron2016M_bin4_ZZtoLep
585 prop_binSR_muon2018_bin7_Triboson
586 prop_binCRWS_muon2017_bin12
587 prop_binCRWS_muon2017_bin9
588 prop_binCRF_electron2016M_bin12_sm
589 prop_binSR_electron2017_bin15_WpWpJJ_QCD
590 prop_binCRF_electron2018_bin5
591 prop_binSR_electron2016M_bin14_TVX
592 prop_binCRTT_electron2017_bin13
593 prop_binCRF_electron2016M_bin18_TTTo2L2Nu
594 prop_binSR_muon2016M_bin19_Triboson
595 prop_binCRTT_electron2016M_bin15
596 prop_binSR_muon2018_bin16_Triboson
597 prop_binSR_muon2016M_bin6_WrongSign
598 prop_binCRF_electron2018_bin15_TTTo2L2Nu
599 prop_binSR_electron2016M_bin7_WrongSign
600 prop_binCRTT_electron2016M_bin6
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

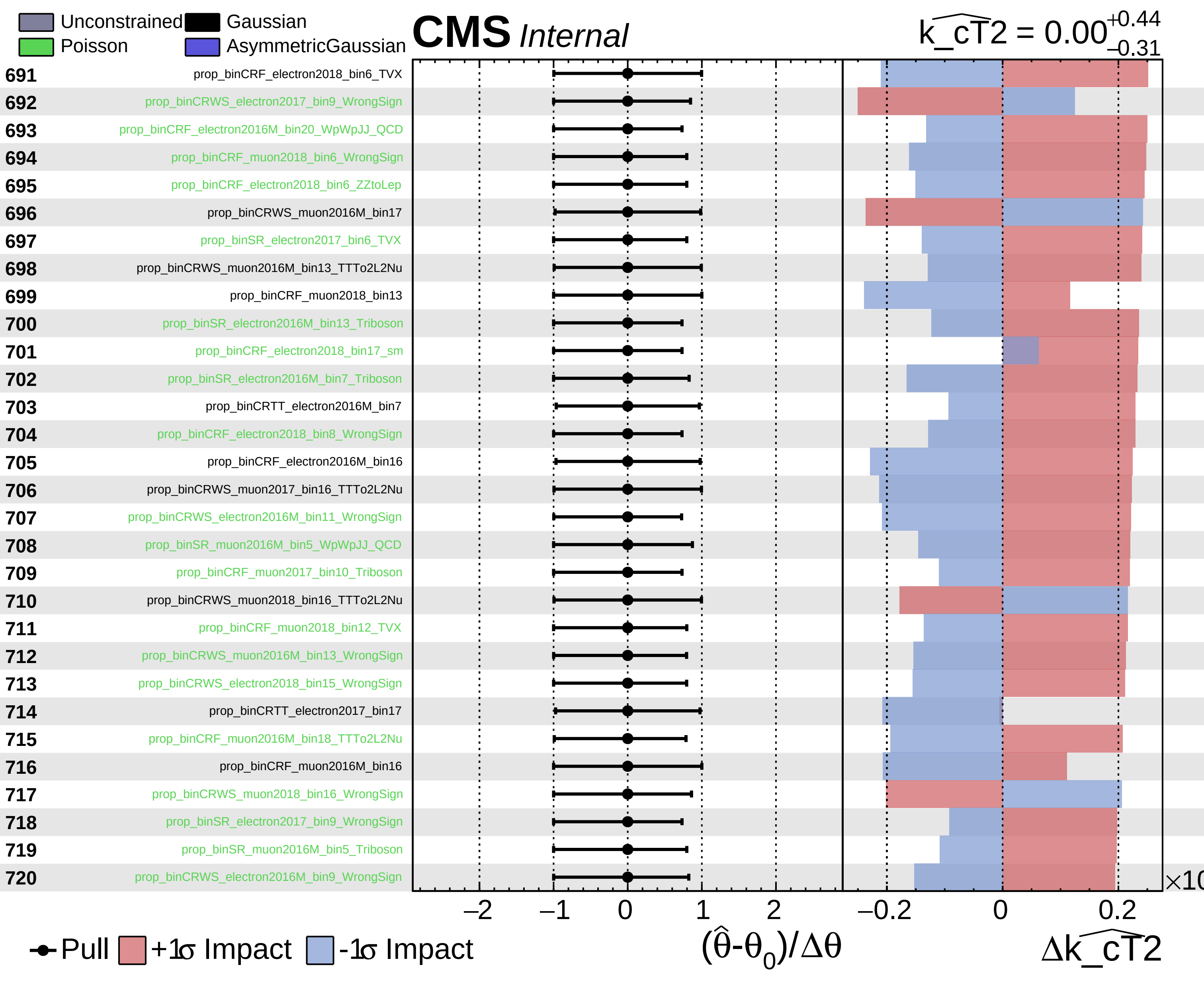


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
601 prop_binCRTT_electron2018_bin15
602 prop_binSR_muon2018_bin16_TVX
603 prop_binSR_muon2017_bin15_Triboson
604 prop_binCRF_muon2017_bin7_TTTo2L2Nu
605 prop_binSR_electron2016M_bin14_WpWpJJ_QCD
606 prop_binCRWS_muon2016M_bin10
607 prop_binCRF_electron2018_bin19
608 prop_binSR_muon2017_bin16_Triboson
609 prop_binSR_muon2018_bin11_Triboson
610 prop_binSR_muon2016M_bin8_Triboson
611 prop_binCRWS_electron2017_bin13
612 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
613 prop_binCRWS_muon2018_bin13
614 prop_binCRF_muon2016M_bin20_sm
615 prop_binCRWS_muon2017_bin20_Fake
616 prop_binCRWS_electron2018_bin13
617 prop_binCRTT_electron2016M_bin13
618 prop_binSR_muon2016M_bin8_TVX
619 prop_binSR_electron2017_bin9_WpWpJJ_QCD
620 prop_binSR_muon2018_bin11_WpWpJJ_QCD
621 prop_binCRF_electron2018_bin17_ZZtoLep
622 prop_binCRF_muon2016M_bin13
623 prop_binCRF_electron2018_bin14
624 prop_binCRWS_electron2016M_bin4_Fake
625 prop_binSR_electron2017_bin11_ZZtoLep
626 prop_binCRF_electron2017_bin5
627 prop_binSR_muon2017_bin12_Triboson
628 prop_binCRF_muon2018_bin7
629 prop_binSR_muon2016M_bin6_WpWpJJ_QCD
630 prop_binCRF_electron2018_bin13
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

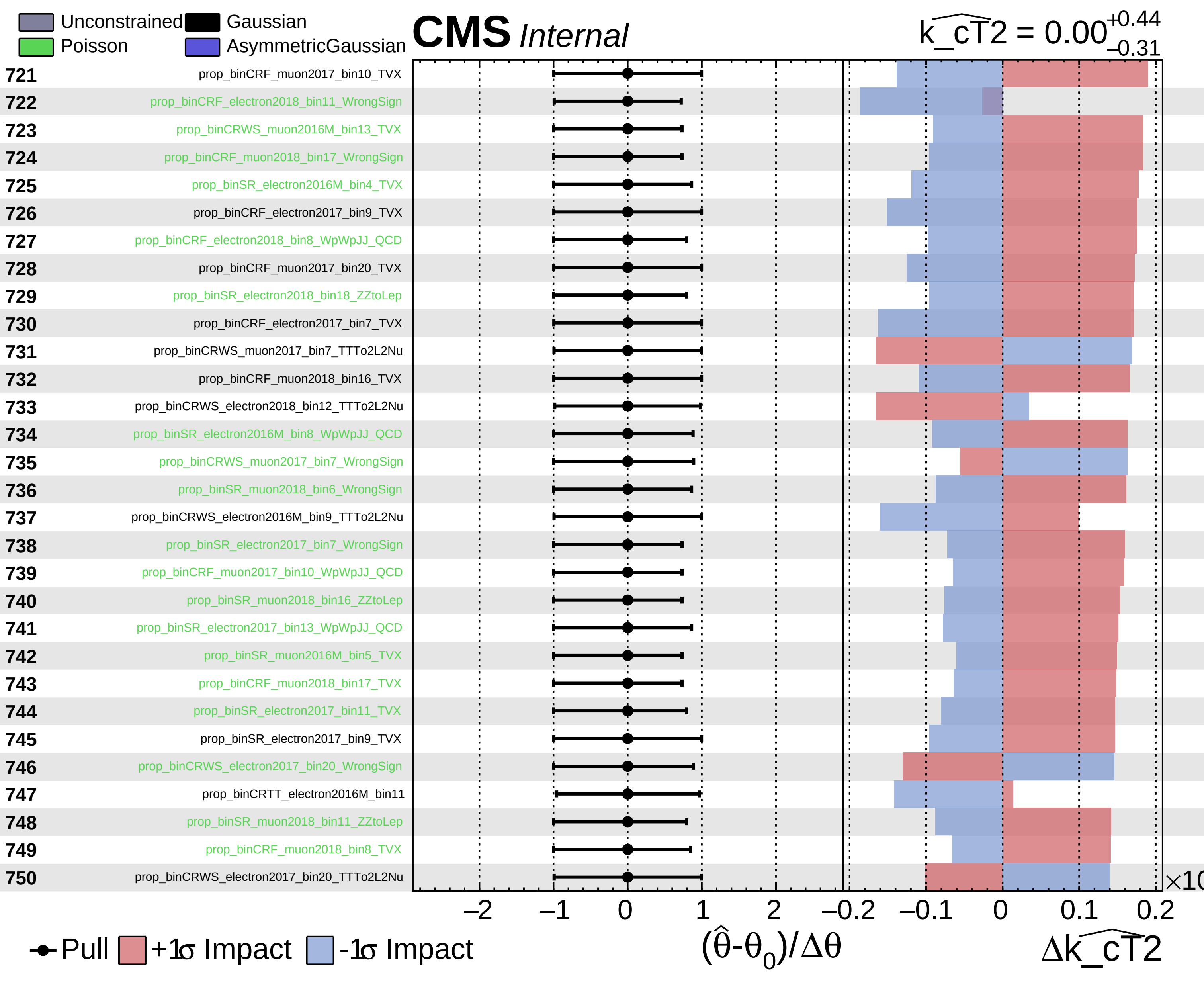


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
631 prop_binCRF_electron2016M_bin6_ZZtoLep
632 prop_binCRF_muon2016M_bin5_sm
633 prop_binSR_electron2016M_bin13_TVX
634 prop_binCRF_electron2016M_bin4
635 prop_binCRF_muon2018_bin8_ZZtoLep
636 prop_binCRF_electron2017_bin6
637 prop_binCRF_electron2018_bin8_TVX
638 prop_binCRF_muon2017_bin20_sm
639 prop_binCRF_muon2018_bin5
640 prop_binCRF_electron2016M_bin8
641 prop_binSR_muon2018_bin7_WrongSign
642 prop_binCRWS_electron2017_bin19
643 prop_binCRWS_muon2017_bin18_Fake
644 prop_binSR_muon2017_bin12_WpWpJJ_QCD
645 prop_binCRF_electron2017_bin13
646 prop_binSR_electron2017_bin9_Triboson
647 prop_binSR_muon2016M_bin19_TVX
648 prop_binSR_muon2017_bin12_TVX
649 prop_binCRF_muon2018_bin8_TTTo2L2Nu
650 prop_binCRWS_electron2016M_bin15
651 prop_binCRF_electron2016M_bin10_sm
652 prop_binCRWS_electron2017_bin18
653 prop_binSR_electron2017_bin6_Triboson
654 prop_binCRF_electron2017_bin11
655 prop_binCRWS_muon2016M_bin13_Fake
656 prop_binCRWS_muon2016M_bin16
657 prop_binSR_electron2017_bin13_TVX
658 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
659 prop_binCRF_electron2016M_bin12_ZZtoLep
660 prop_binSR_electron2016M_bin14_Triboson
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

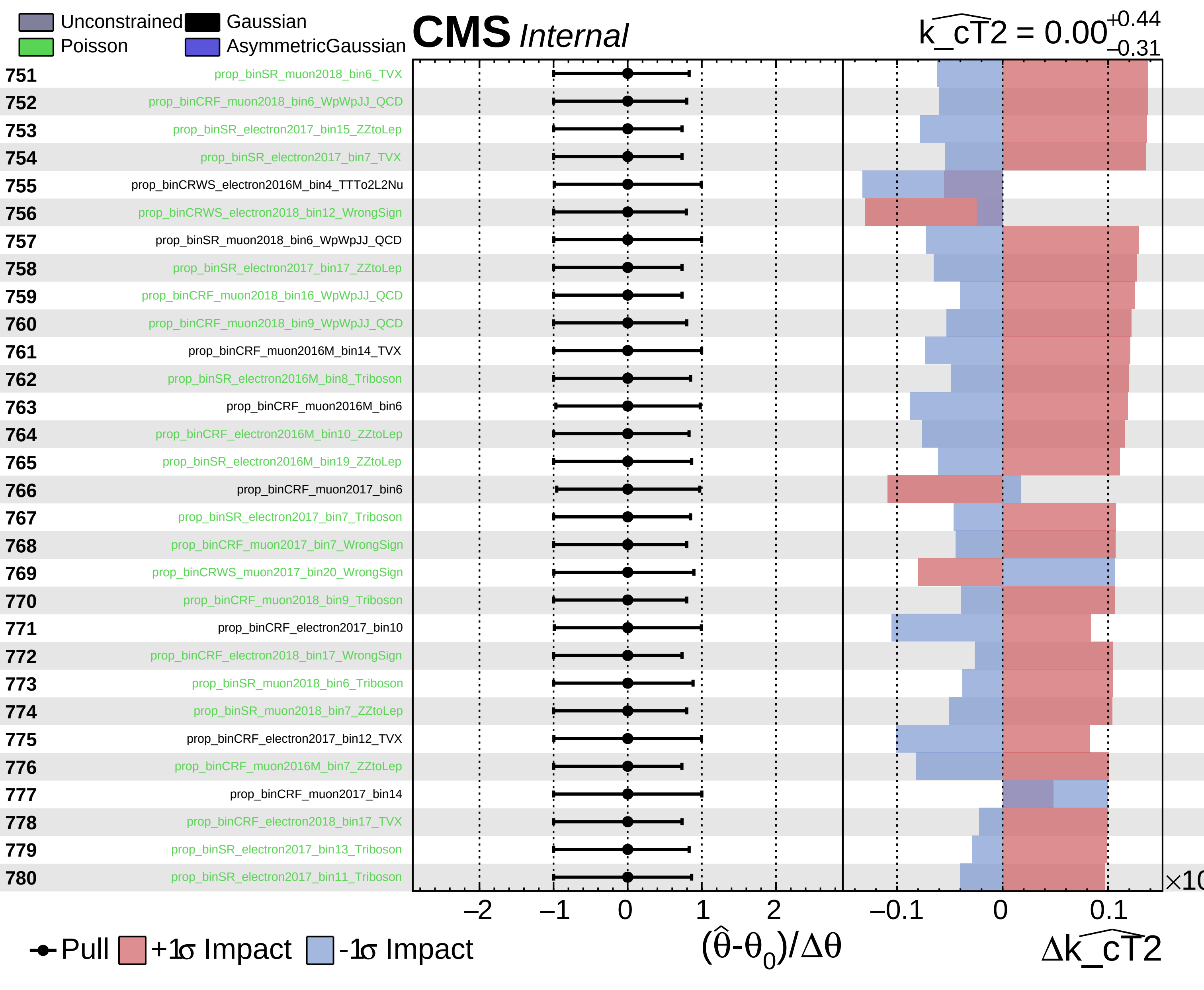

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
661 prop_binCRWS_muon2017_bin19
662 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
663 prop_binSR_muon2017_bin14_WpWpJJ_QCD
664 prop_binSR_muon2016M_bin6_TVX
665 prop_binCRWS_muon2018_bin11
666 prop_binCRF_muon2016M_bin11
667 prop_binCRF_electron2016M_bin17
668 prop_binCRWS_electron2017_bin8
669 prop_binCRWS_muon2016M_bin4_TTTo2L2Nu
670 prop_binCRF_muon2018_bin6_TVX
671 prop_binCRF_muon2016M_bin20_WpWpJJ_QCD
672 prop_binCRWS_muon2017_bin6
673 prop_binCRF_muon2016M_bin5_TTTo2L2Nu
674 prop_binSR_muon2016M_bin16_Triboson
675 prop_binSR_electron2016M_bin8_TVX
676 prop_binCRF_muon2016M_bin9
677 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
678 prop_binSR_electron2017_bin17_TVX
679 prop_binCRF_electron2016M_bin10_TTTo2L2Nu
680 prop_binSR_muon2016M_bin6_Triboson
681 prop_binCRWS_electron2018_bin15_TTTo2L2Nu
682 prop_binCRF_electron2016M_bin20_Triboson
683 prop_binCRWS_muon2017_bin16_WrongSign
684 prop_binSR_electron2017_bin7_WpWpJJ_QCD
685 prop_binCRWS_muon2016M_bin4_WrongSign
686 prop_binCRF_muon2018_bin9_TVX
687 prop_binSR_muon2017_bin14_Triboson
688 prop_binCRF_electron2016M_bin7
689 prop_binCRTT_electron2017_bin11
690 prop_binCRF_electron2017_bin15
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
691 prop_binCRF_electron2018_bin6_TVX
692 prop_binCRWS_electron2017_bin9_WrongSign
693 prop_binCRF_electron2016M_bin20_WpWpJJ_QCD
694 prop_binCRF_muon2018_bin6_WrongSign
695 prop_binCRF_electron2018_bin6_ZZtoLep
696 prop_binCRWS_muon2016M_bin17
697 prop_binSR_electron2017_bin6_TVX
698 prop_binCRWS_muon2016M_bin13_TTTo2L2Nu
699 prop_binCRF_muon2018_bin13
700 prop_binSR_electron2016M_bin13_Triboson
701 prop_binCRF_electron2018_bin17_sm
702 prop_binSR_electron2016M_bin7_Triboson
703 prop_binCRTT_electron2016M_bin7
704 prop_binCRF_electron2018_bin8_WrongSign
705 prop_binCRF_electron2016M_bin16
706 prop_binCRWS_muon2017_bin16_TTTo2L2Nu
707 prop_binCRWS_electron2016M_bin11_WrongSign
708 prop_binSR_muon2016M_bin5_WpWpJJ_QCD
709 prop_binCRF_muon2017_bin10_Triboson
710 prop_binCRWS_muon2018_bin16_TTTo2L2Nu
711 prop_binCRF_muon2018_bin12_TVX
712 prop_binCRWS_muon2016M_bin13_WrongSign
713 prop_binCRWS_electron2018_bin15_WrongSign
714 prop_binCRTT_electron2017_bin17
715 prop_binCRF_muon2016M_bin18_TTTo2L2Nu
716 prop_binCRF_muon2016M_bin16
717 prop_binCRWS_muon2018_bin16_WrongSign
718 prop_binSR_electron2017_bin9_WrongSign
719 prop_binSR_muon2016M_bin5_Triboson
720 prop_binCRWS_electron2016M_bin9_WrongSign
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
721 prop_binCRF_muon2017_bin10_TVX
722 prop_binCRF_electron2018_bin11_WrongSign
723 prop_binCRWS_muon2016M_bin13_TVX
724 prop_binCRF_muon2018_bin17_WrongSign
725 prop_binSR_electron2016M_bin4_TVX
726 prop_binCRF_electron2017_bin9_TVX
727 prop_binCRF_electron2018_bin8_WpWpJJ_QCD
728 prop_binCRF_muon2017_bin20_TVX
729 prop_binSR_electron2018_bin18_ZZtoLep
730 prop_binCRF_electron2017_bin7_TVX
731 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
732 prop_binCRF_muon2018_bin16_TVX
733 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
734 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
735 prop_binCRWS_muon2017_bin7_WrongSign
736 prop_binSR_muon2018_bin6_WrongSign
737 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
738 prop_binSR_electron2017_bin7_WrongSign
739 prop_binCRF_muon2017_bin10_WpWpJJ_QCD
740 prop_binSR_muon2018_bin16_ZZtoLep
741 prop_binSR_electron2017_bin13_WpWpJJ_QCD
742 prop_binSR_muon2016M_bin5_TVX
743 prop_binCRF_muon2018_bin17_TVX
744 prop_binSR_electron2017_bin11_TVX
745 prop_binSR_electron2017_bin9_TVX
746 prop_binCRWS_electron2017_bin20_WrongSign
747 prop_binCRTT_electron2016M_bin11
748 prop_binSR_muon2018_bin11_ZZtoLep
749 prop_binCRF_muon2018_bin8_TVX
750 prop_binCRWS_electron2017_bin20_TTTo2L2Nu
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2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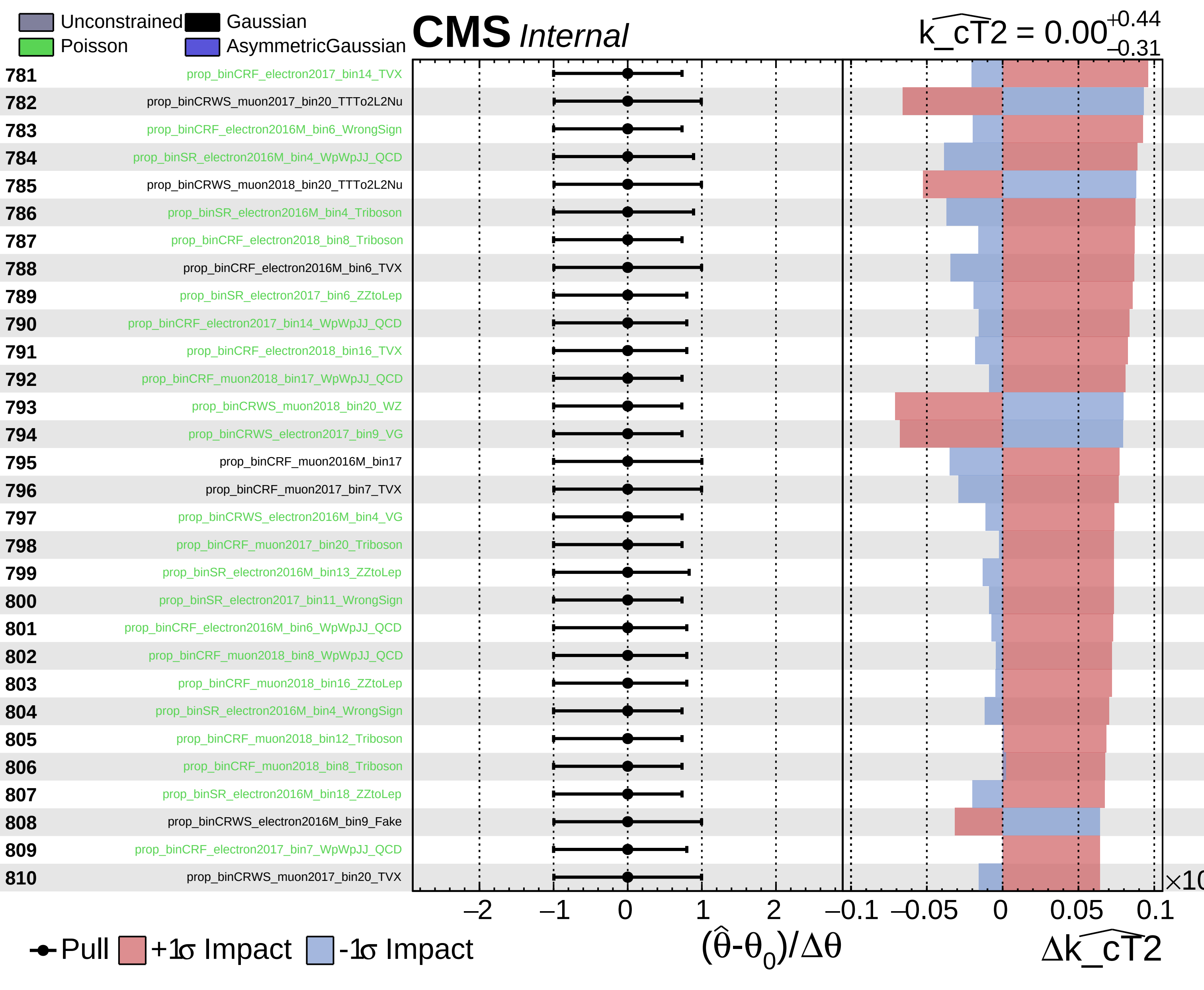
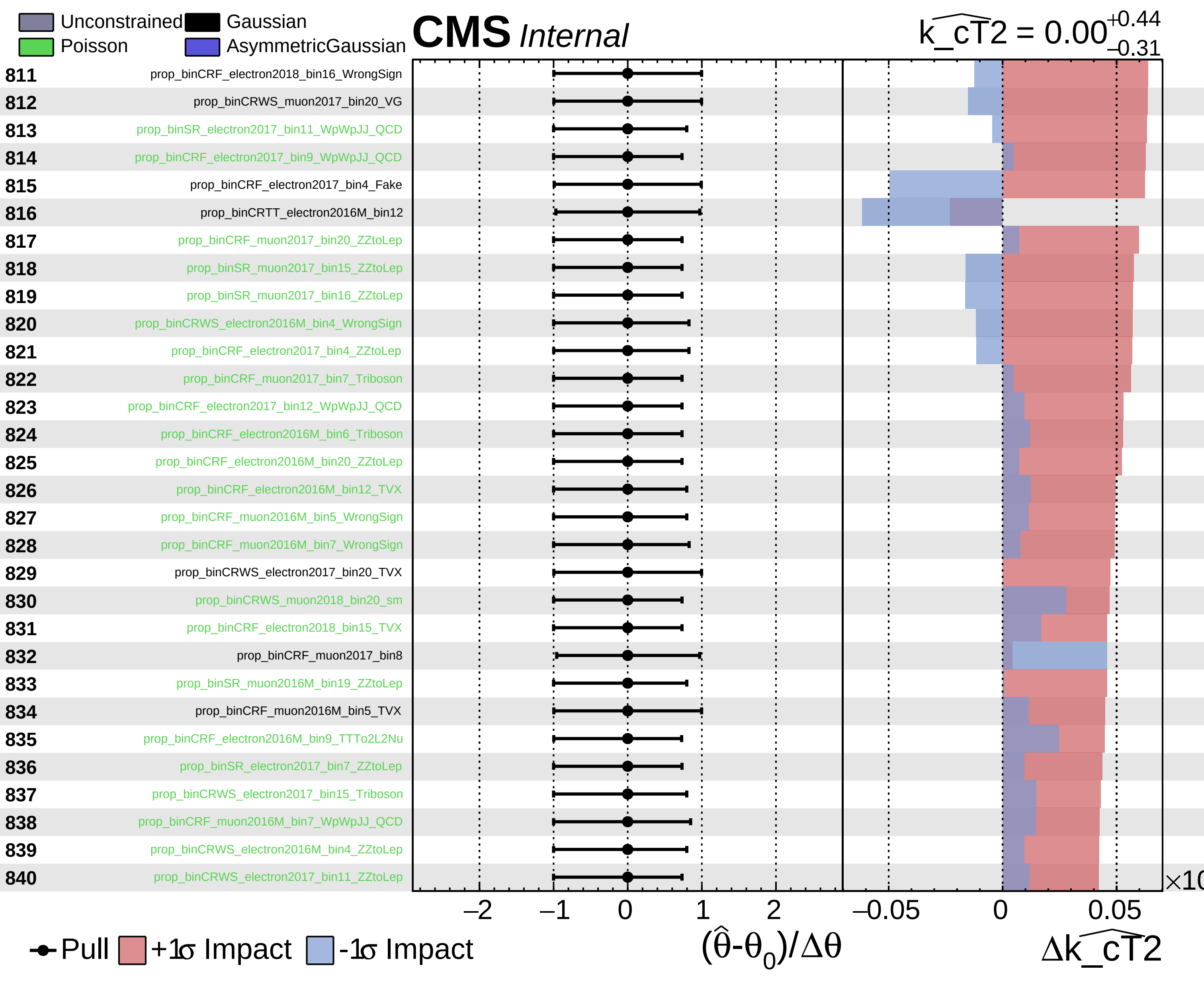
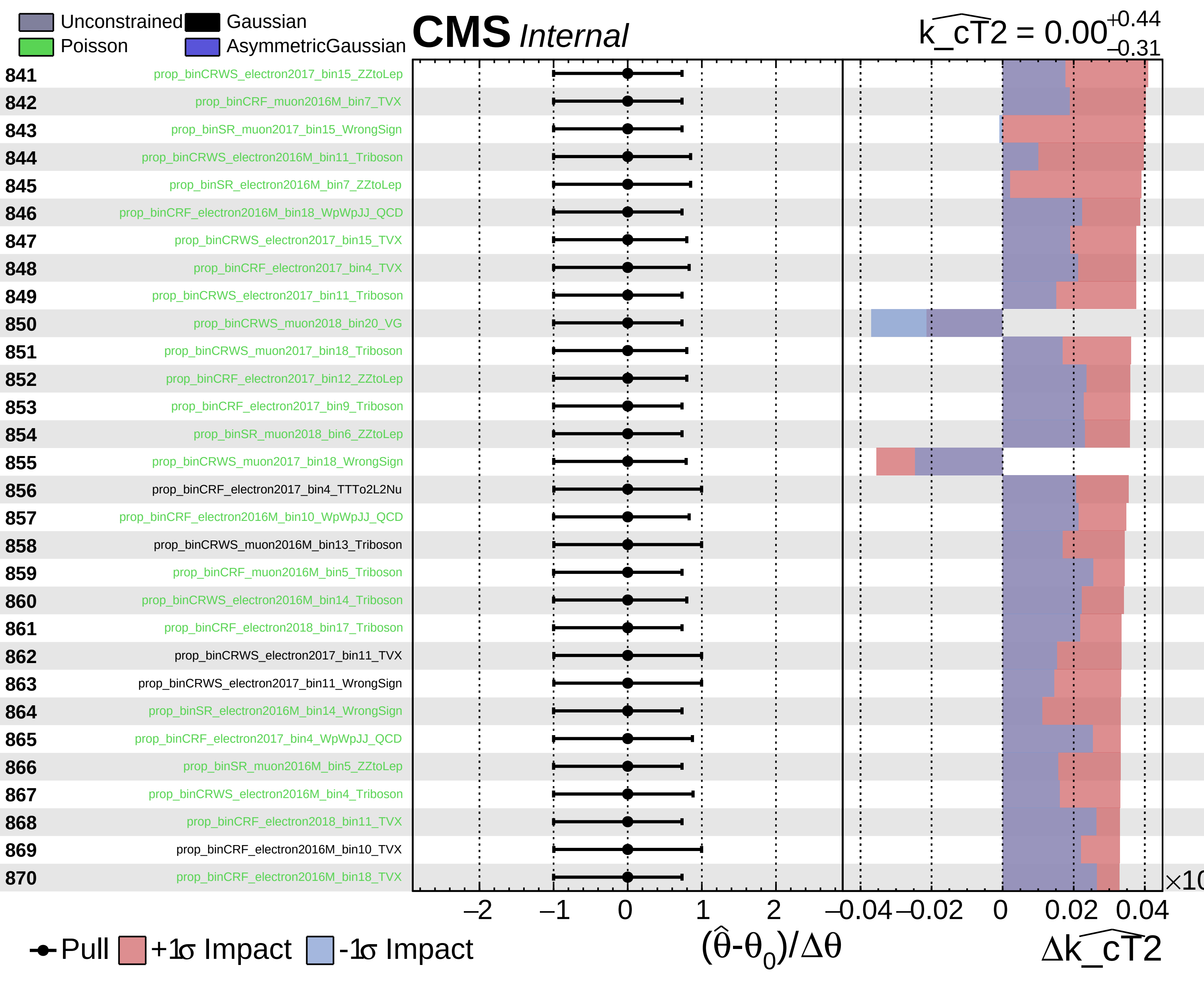
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
751 prop_binSR_muon2018_bin6_TVX
752 prop_binCRF_muon2018_bin6_WpWpJJ_QCD
753 prop_binSR_electron2017_bin15_ZZtoLep
754 prop_binSR_electron2017_bin7_TVX
755 prop_binCRWS_electron2016M_bin4_TTTo2L2Nu
756 prop_binCRWS_electron2018_bin12_WrongSign
757 prop_binSR_muon2018_bin6_WpWpJJ_QCD
758 prop_binSR_electron2017_bin17_ZZtoLep
759 prop_binCRF_muon2018_bin16_WpWpJJ_QCD
760 prop_binCRF_muon2018_bin9_WpWpJJ_QCD
761 prop_binCRF_muon2016M_bin14_TVX
762 prop_binSR_electron2016M_bin8_Triboson
763 prop_binCRF_muon2016M_bin6
764 prop_binCRF_electron2016M_bin10_ZZtoLep
765 prop_binSR_electron2016M_bin19_ZZtoLep
766 prop_binCRF_muon2017_bin6
767 prop_binSR_electron2017_bin7_Triboson
768 prop_binCRF_muon2017_bin7_WrongSign
769 prop_binCRWS_muon2017_bin20_WrongSign
770 prop_binCRF_muon2018_bin9_Triboson
771 prop_binCRF_electron2017_bin10
772 prop_binCRF_electron2018_bin17_WrongSign
773 prop_binSR_muon2018_bin6_Triboson
774 prop_binSR_muon2018_bin7_ZZtoLep
775 prop_binCRF_electron2017_bin12_TVX
776 prop_binCRF_muon2016M_bin7_ZZtoLep
777 prop_binCRF_muon2017_bin14
778 prop_binCRF_electron2018_bin17_TVX
779 prop_binSR_electron2017_bin13_Triboson
780 prop_binSR_electron2017_bin11_Triboson
×10
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

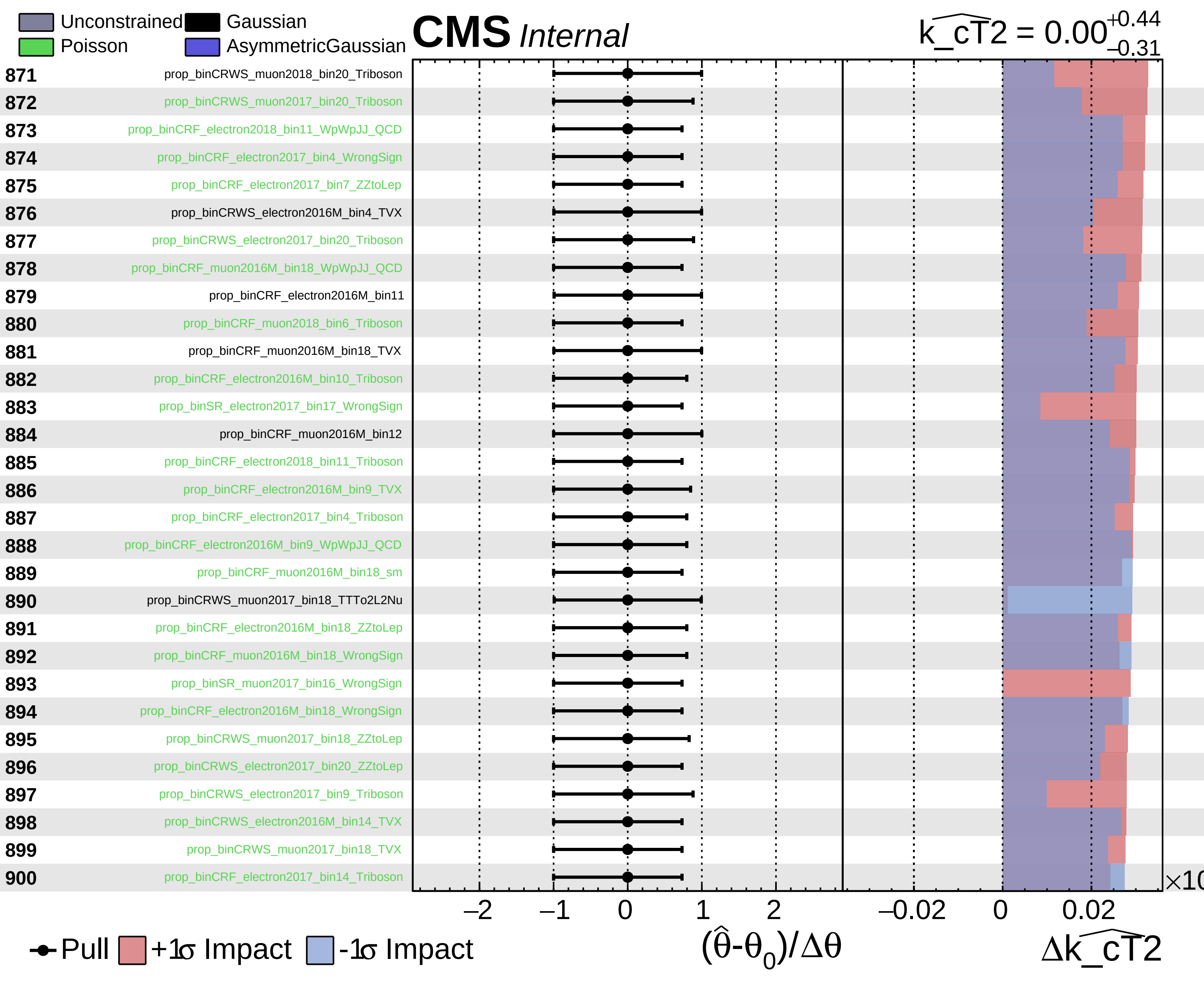


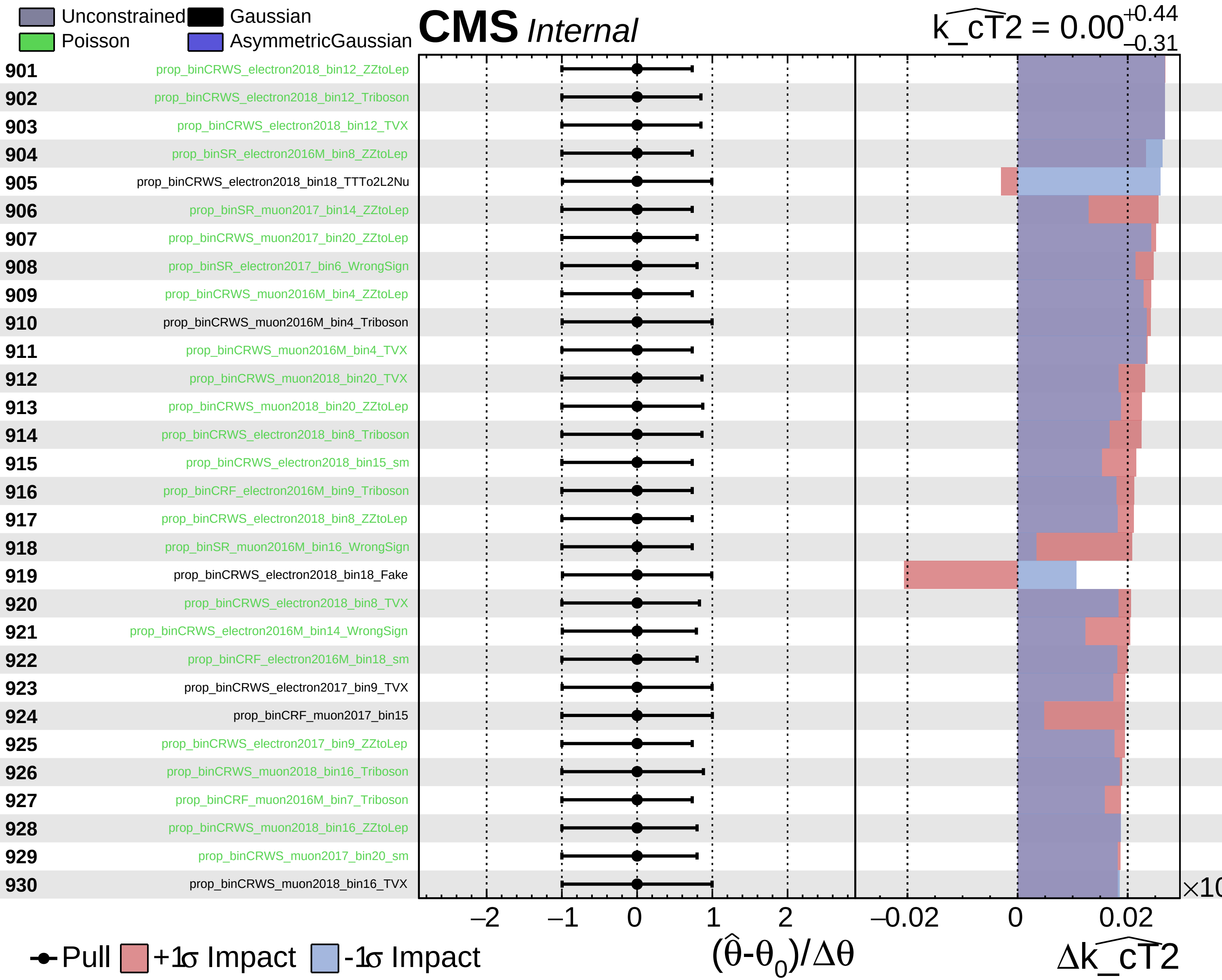
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
781 prop_binCRF_electron2017_bin14_TVX
782 prop_binCRWS_muon2017_bin20_TTTo2L2Nu
783 prop_binCRF_electron2016M_bin6_WrongSign
784 prop_binSR_electron2016M_bin4_WpWpJJ_QCD
785 prop_binCRWS_muon2018_bin20_TTTo2L2Nu
786 prop_binSR_electron2016M_bin4_Triboson
787 prop_binCRF_electron2018_bin8_Triboson
788 prop_binCRF_electron2016M_bin6_TVX
789 prop_binSR_electron2017_bin6_ZZtoLep
790 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
791 prop_binCRF_electron2018_bin16_TVX
792 prop_binCRF_muon2018_bin17_WpWpJJ_QCD
793 prop_binCRWS_muon2018_bin20_WZ
794 prop_binCRWS_electron2017_bin9_VG
795 prop_binCRF_muon2016M_bin17
796 prop_binCRF_muon2017_bin7_TVX
797 prop_binCRWS_electron2016M_bin4_VG
798 prop_binCRF_muon2017_bin20_Triboson
799 prop_binSR_electron2016M_bin13_ZZtoLep
800 prop_binSR_electron2017_bin11_WrongSign
801 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
802 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
803 prop_binCRF_muon2018_bin16_ZZtoLep
804 prop_binSR_electron2016M_bin4_WrongSign
805 prop_binCRF_muon2018_bin12_Triboson
806 prop_binCRF_muon2018_bin8_Triboson
807 prop_binSR_electron2016M_bin18_ZZtoLep
808 prop_binCRWS_electron2016M_bin9_Fake
809 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
810 prop_binCRWS_muon2017_bin20_TVX
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

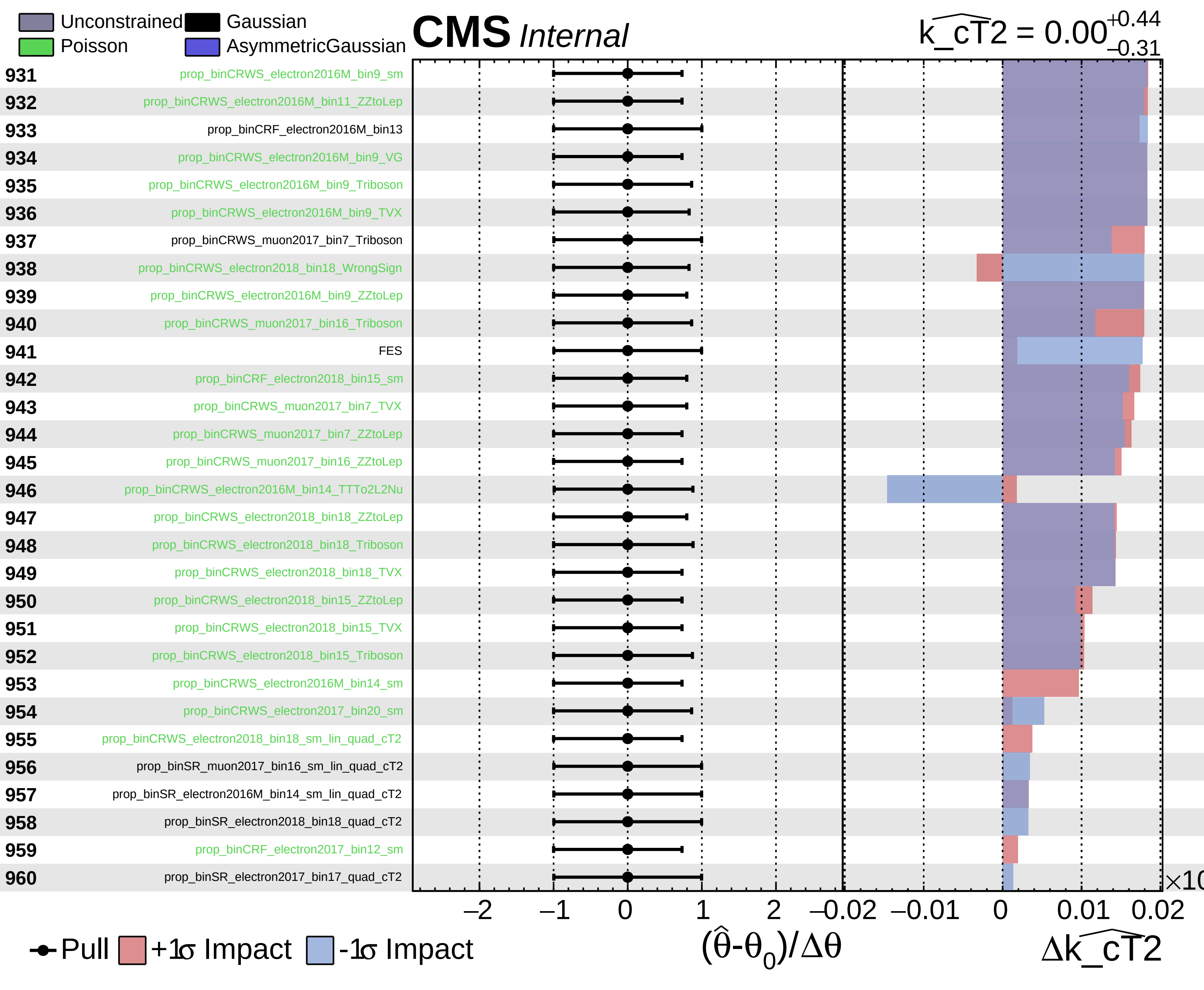

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
811 prop_binCRF_electron2018_bin16_WrongSign
812 prop_binCRWS_muon2017_bin20_VG
813 prop_binSR_electron2017_bin11_WpWpJJ_QCD
814 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
815 prop_binCRF_electron2017_bin4_Fake
816 prop_binCRTT_electron2016M_bin12
817 prop_binCRF_muon2017_bin20_ZZtoLep
818 prop_binSR_muon2017_bin15_ZZtoLep
819 prop_binSR_muon2017_bin16_ZZtoLep
820 prop_binCRWS_electron2016M_bin4_WrongSign
821 prop_binCRF_electron2017_bin4_ZZtoLep
822 prop_binCRF_muon2017_bin7_Triboson
823 prop_binCRF_electron2017_bin12_WpWpJJ_QCD
824 prop_binCRF_electron2016M_bin6_Triboson
825 prop_binCRF_electron2016M_bin20_ZZtoLep
826 prop_binCRF_electron2016M_bin12_TVX
827 prop_binCRF_muon2016M_bin5_WrongSign
828 prop_binCRF_muon2016M_bin7_WrongSign
829 prop_binCRWS_electron2017_bin20_TVX
830 prop_binCRWS_muon2018_bin20_sm
831 prop_binCRF_electron2018_bin15_TVX
832 prop_binCRF_muon2017_bin8
833 prop_binSR_muon2016M_bin19_ZZtoLep
834 prop_binCRF_muon2016M_bin5_TVX
835 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
836 prop_binSR_electron2017_bin7_ZZtoLep
837 prop_binCRWS_electron2017_bin15_Triboson
838 prop_binCRF_muon2016M_bin7_WpWpJJ_QCD
839 prop_binCRWS_electron2016M_bin4_ZZtoLep
840 prop_binCRWS_electron2017_bin11_ZZtoLep
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



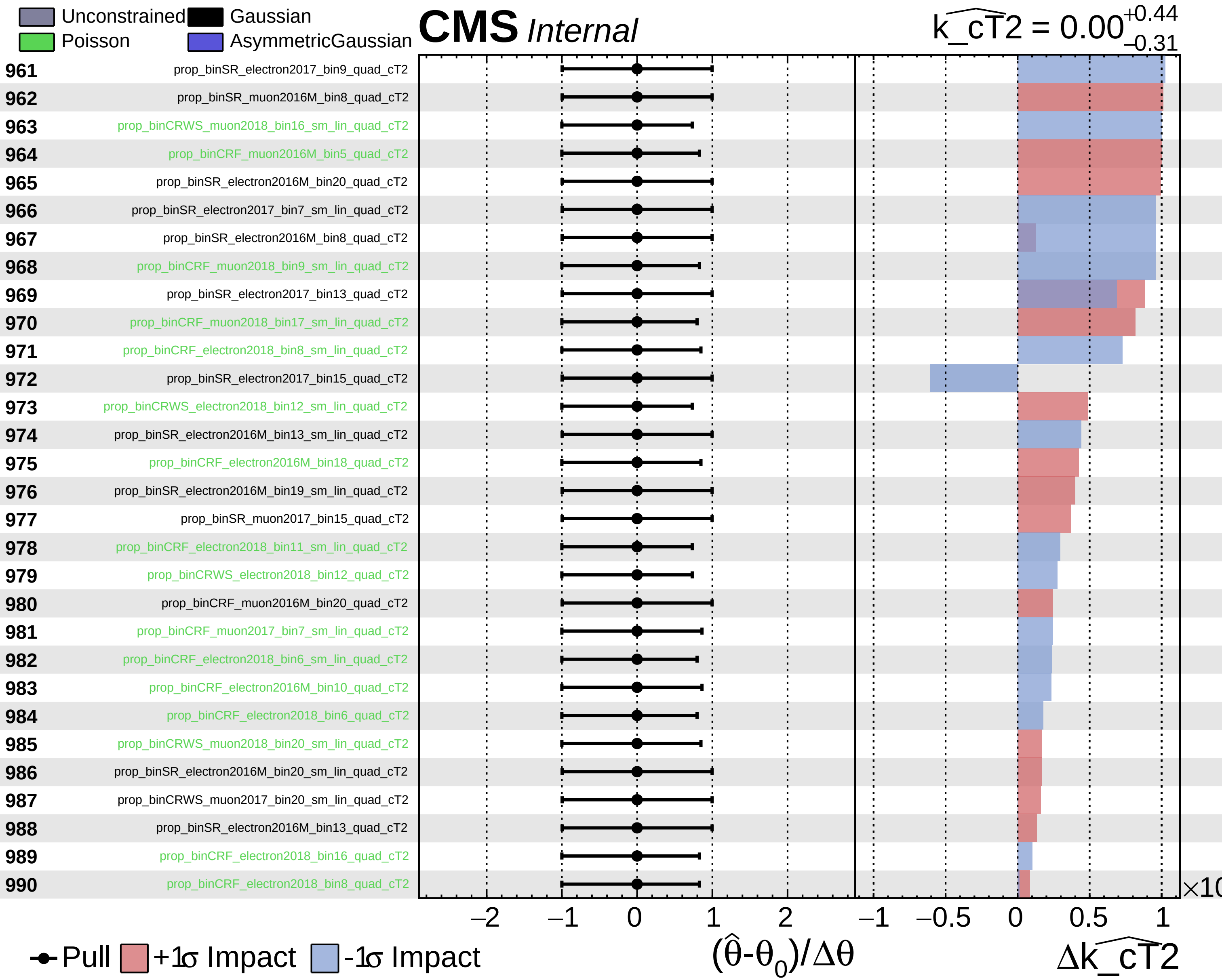
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
841 prop_binCRWS_electron2017_bin15_ZZtoLep
842 prop_binCRF_muon2016M_bin7_TVX
843 prop_binSR_muon2017_bin15_WrongSign
844 prop_binCRWS_electron2016M_bin11_Triboson
845 prop_binSR_electron2016M_bin7_ZZtoLep
846 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
847 prop_binCRWS_electron2017_bin15_TVX
848 prop_binCRF_electron2017_bin4_TVX
849 prop_binCRWS_electron2017_bin11_Triboson
850 prop_binCRWS_muon2018_bin20_VG
851 prop_binCRWS_muon2017_bin18_Triboson
852 prop_binCRF_electron2017_bin12_ZZtoLep
853 prop_binCRF_electron2017_bin9_Triboson
854 prop_binSR_muon2018_bin6_ZZtoLep
855 prop_binCRWS_muon2017_bin18_WrongSign
856 prop_binCRF_electron2017_bin4_TTTo2L2Nu
857 prop_binCRF_electron2016M_bin10_WpWpJJ_QCD
858 prop_binCRWS_muon2016M_bin13_Triboson
859 prop_binCRF_muon2016M_bin5_Triboson
860 prop_binCRWS_electron2016M_bin14_Triboson
861 prop_binCRF_electron2018_bin17_Triboson
862 prop_binCRWS_electron2017_bin11_TVX
863 prop_binCRWS_electron2017_bin11_WrongSign
864 prop_binSR_electron2016M_bin14_WrongSign
865 prop_binCRF_electron2017_bin4_WpWpJJ_QCD
866 prop_binSR_muon2016M_bin5_ZZtoLep
867 prop_binCRWS_electron2016M_bin4_Triboson
868 prop_binCRF_electron2018_bin11_TVX
869 prop_binCRF_electron2016M_bin10_TVX
870 prop_binCRF_electron2016M_bin18_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
871 prop_binCRWS_muon2018_bin20_Triboson
872 prop_binCRWS_muon2017_bin20_Triboson
873 prop_binCRF_electron2018_bin11_WpWpJJ_QCD
874 prop_binCRF_electron2017_bin4_WrongSign
875 prop_binCRF_electron2017_bin7_ZZtoLep
876 prop_binCRWS_electron2016M_bin4_TVX
877 prop_binCRWS_electron2017_bin20_Triboson
878 prop_binCRF_muon2016M_bin18_WpWpJJ_QCD
879 prop_binCRF_electron2016M_bin11
880 prop_binCRF_muon2018_bin6_Triboson
881 prop_binCRF_muon2016M_bin18_TVX
882 prop_binCRF_electron2016M_bin10_Triboson
883 prop_binSR_electron2017_bin17_WrongSign
884 prop_binCRF_muon2016M_bin12
885 prop_binCRF_electron2018_bin11_Triboson
886 prop_binCRF_electron2016M_bin9_TVX
887 prop_binCRF_electron2017_bin4_Triboson
888 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
889 prop_binCRF_muon2016M_bin18_sm
890 prop_binCRWS_muon2017_bin18_TTTo2L2Nu
891 prop_binCRF_electron2016M_bin18_ZZtoLep
892 prop_binCRF_muon2016M_bin18_WrongSign
893 prop_binSR_muon2017_bin16_WrongSign
894 prop_binCRF_electron2016M_bin18_WrongSign
895 prop_binCRWS_muon2017_bin18_ZZtoLep
896 prop_binCRWS_electron2017_bin20_ZZtoLep
897 prop_binCRWS_electron2017_bin9_Triboson
898 prop_binCRWS_electron2016M_bin14_TVX
899 prop_binCRWS_muon2017_bin18_TVX
900 prop_binCRF_electron2017_bin14_Triboson
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
901 prop_binCRWS_electron2018_bin12_ZZtoLep
902 prop_binCRWS_electron2018_bin12_Triboson
903 prop_binCRWS_electron2018_bin12_TVX
904 prop_binSR_electron2016M_bin8_ZZtoLep
905 prop_binCRWS_electron2018_bin18_TTTo2L2Nu
906 prop_binSR_muon2017_bin14_ZZtoLep
907 prop_binCRWS_muon2017_bin20_ZZtoLep
908 prop_binSR_electron2017_bin6_WrongSign
909 prop_binCRWS_muon2016M_bin4_ZZtoLep
910 prop_binCRWS_muon2016M_bin4_Triboson
911 prop_binCRWS_muon2016M_bin4_TVX
912 prop_binCRWS_muon2018_bin20_TVX
913 prop_binCRWS_muon2018_bin20_ZZtoLep
914 prop_binCRWS_electron2018_bin8_Triboson
915 prop_binCRWS_electron2018_bin15_sm
916 prop_binCRF_electron2016M_bin9_Triboson
917 prop_binCRWS_electron2018_bin8_ZZtoLep
918 prop_binSR_muon2016M_bin16_WrongSign
919 prop_binCRWS_electron2018_bin18_Fake
920 prop_binCRWS_electron2018_bin8_TVX
921 prop_binCRWS_electron2016M_bin14_WrongSign
922 prop_binCRF_electron2016M_bin18_sm
923 prop_binCRWS_electron2017_bin9_TVX
924 prop_binCRF_muon2017_bin15
925 prop_binCRWS_electron2017_bin9_ZZtoLep
926 prop_binCRWS_muon2018_bin16_Triboson
927 prop_binCRF_muon2016M_bin7_Triboson
928 prop_binCRWS_muon2018_bin16_ZZtoLep
929 prop_binCRWS_muon2017_bin20_sm
930 prop_binCRWS_muon2018_bin16_TVX
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
931 prop_binCRWS_electron2016M_bin9_sm
932 prop_binCRWS_electron2016M_bin11_ZZtoLep
933 prop_binCRF_electron2016M_bin13
934 prop_binCRWS_electron2016M_bin9_VG
935 prop_binCRWS_electron2016M_bin9_Triboson
936 prop_binCRWS_electron2016M_bin9_TVX
937 prop_binCRWS_muon2017_bin7_Triboson
938 prop_binCRWS_electron2018_bin18_WrongSign
939 prop_binCRWS_electron2016M_bin9_ZZtoLep
940 prop_binCRWS_muon2017_bin16_Triboson
941 FES
942 prop_binCRF_electron2018_bin15_sm
943 prop_binCRWS_muon2017_bin7_TVX
944 prop_binCRWS_muon2017_bin7_ZZtoLep
945 prop_binCRWS_muon2017_bin16_ZZtoLep
946 prop_binCRWS_electron2016M_bin14_TTTo2L2Nu
947 prop_binCRWS_electron2018_bin18_ZZtoLep
948 prop_binCRWS_electron2018_bin18_Triboson
949 prop_binCRWS_electron2018_bin18_TVX
950 prop_binCRWS_electron2018_bin15_ZZtoLep
951 prop_binCRWS_electron2018_bin15_TVX
952 prop_binCRWS_electron2018_bin15_Triboson
953 prop_binCRWS_electron2016M_bin14_sm
954 prop_binCRWS_electron2017_bin20_sm
955 prop_binCRWS_electron2018_bin18_sm_lin_quad_cT2
956 prop_binSR_muon2017_bin16_sm_lin_quad_cT2
957 prop_binSR_electron2016M_bin14_sm_lin_quad_cT2
958 prop_binSR_electron2018_bin18_quad_cT2
959 prop_binCRF_electron2017_bin12_sm
960 prop_binSR_electron2017_bin17_quad_cT2
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
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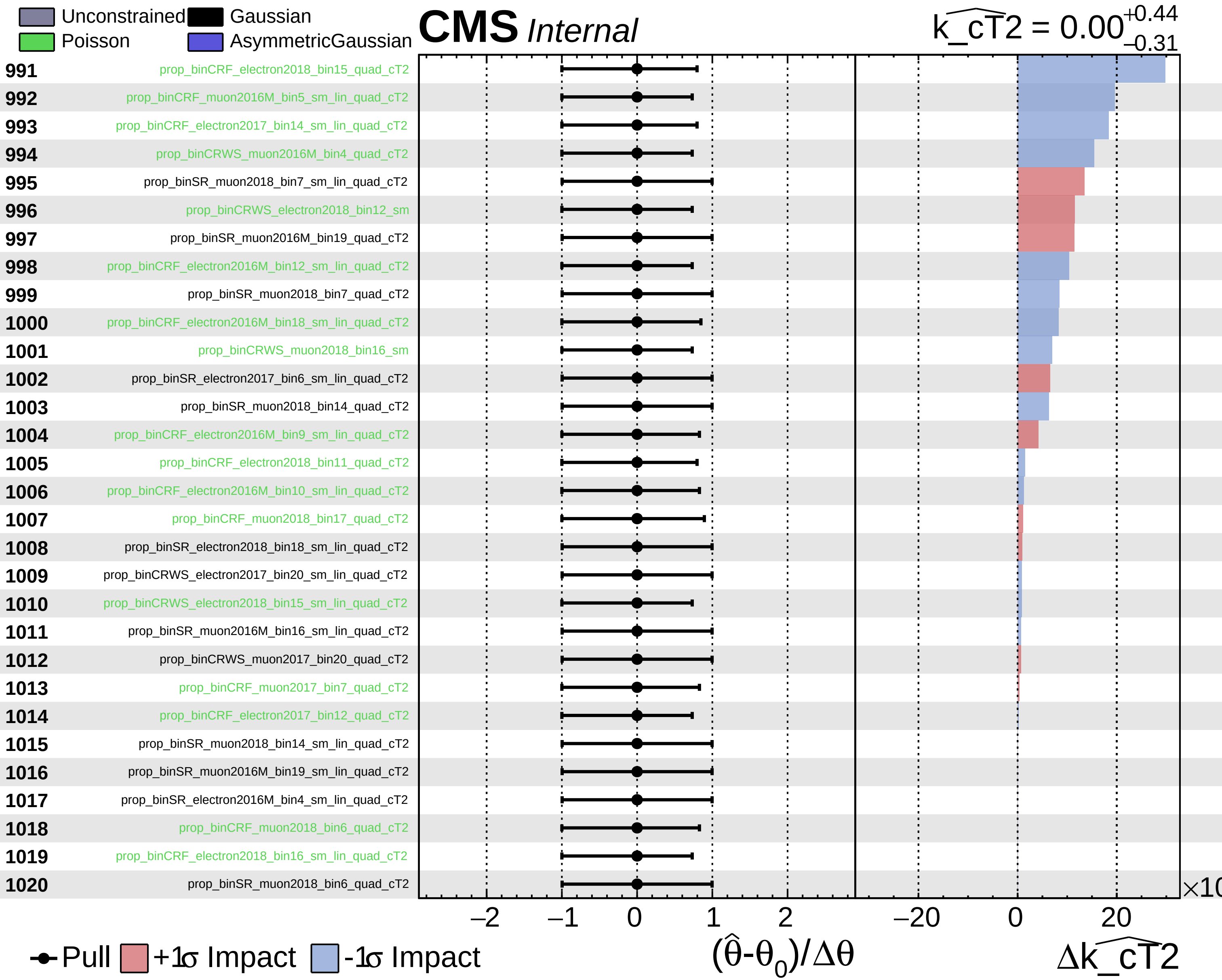
962 prop_binSR_muon2016M_bin8_quad_cT2
963 prop_binCRWS_muon2018_bin16_sm_lin_quad_cT2
964 prop_binCRF_muon2016M_bin5_quad_cT2
965 prop_binSR_electron2016M_bin20_quad_cT2
966 prop_binSR_electron2017_bin7_sm_lin_quad_cT2
967 prop_binSR_electron2016M_bin8_quad_cT2
968 prop_binCRF_muon2018_bin9_sm_lin_quad_cT2
969 prop_binSR_electron2017_bin13_quad_cT2
970 prop_binCRF_muon2018_bin17_sm_lin_quad_cT2
971 prop_binCRF_electron2018_bin8_sm_lin_quad_cT2
972 prop_binSR_electron2017_bin15_quad_cT2
973 prop_binCRWS_electron2018_bin12_sm_lin_quad_cT2
974 prop_binSR_electron2016M_bin13_sm_lin_quad_cT2
975 prop_binCRF_electron2016M_bin18_quad_cT2
976 prop_binSR_electron2016M_bin19_sm_lin_quad_cT2
977 prop_binSR_muon2017_bin15_quad_cT2
978 prop_binCRF_electron2018_bin11_sm_lin_quad_cT2
979 prop_binCRWS_electron2018_bin12_quad_cT2
980 prop_binCRF_muon2016M_bin20_quad_cT2
981 prop_binCRF_muon2017_bin7_sm_lin_quad_cT2
982 prop_binCRF_electron2018_bin6_sm_lin_quad_cT2
983 prop_binCRF_electron2016M_bin10_quad_cT2
984 prop_binCRF_electron2018_bin6_quad_cT2
985 prop_binCRWS_muon2018_bin20_sm_lin_quad_cT2
986 prop_binSR_electron2016M_bin20_sm_lin_quad_cT2
987 prop_binCRWS_muon2017_bin20_sm_lin_quad_cT2
988 prop_binSR_electron2016M_bin13_quad_cT2
989 prop_binCRF_electron2018_bin16_quad_cT2
990 prop_binCRF_electron2018_bin8_quad_cT2
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
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992 prop_binCRF_muon2016M_bin5_sm_lin_quad_cT2
993 prop_binCRF_electron2017_bin14_sm_lin_quad_cT2
994 prop_binCRWS_muon2016M_bin4_quad_cT2
995 prop_binSR_muon2018_bin7_sm_lin_quad_cT2
996 prop_binCRWS_electron2018_bin12_sm
997 prop_binSR_muon2016M_bin19_quad_cT2
998 prop_binCRF_electron2016M_bin12_sm_lin_quad_cT2
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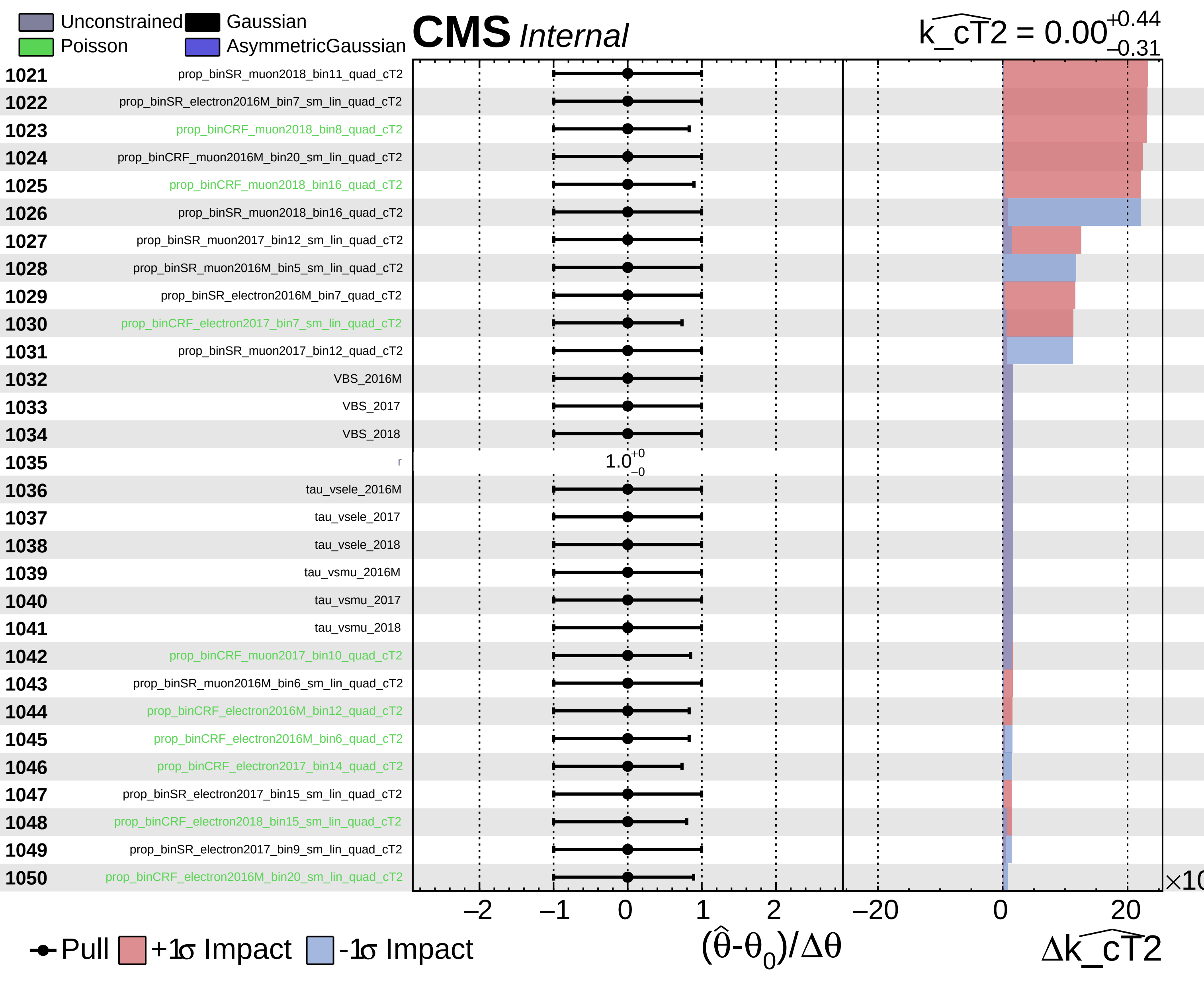
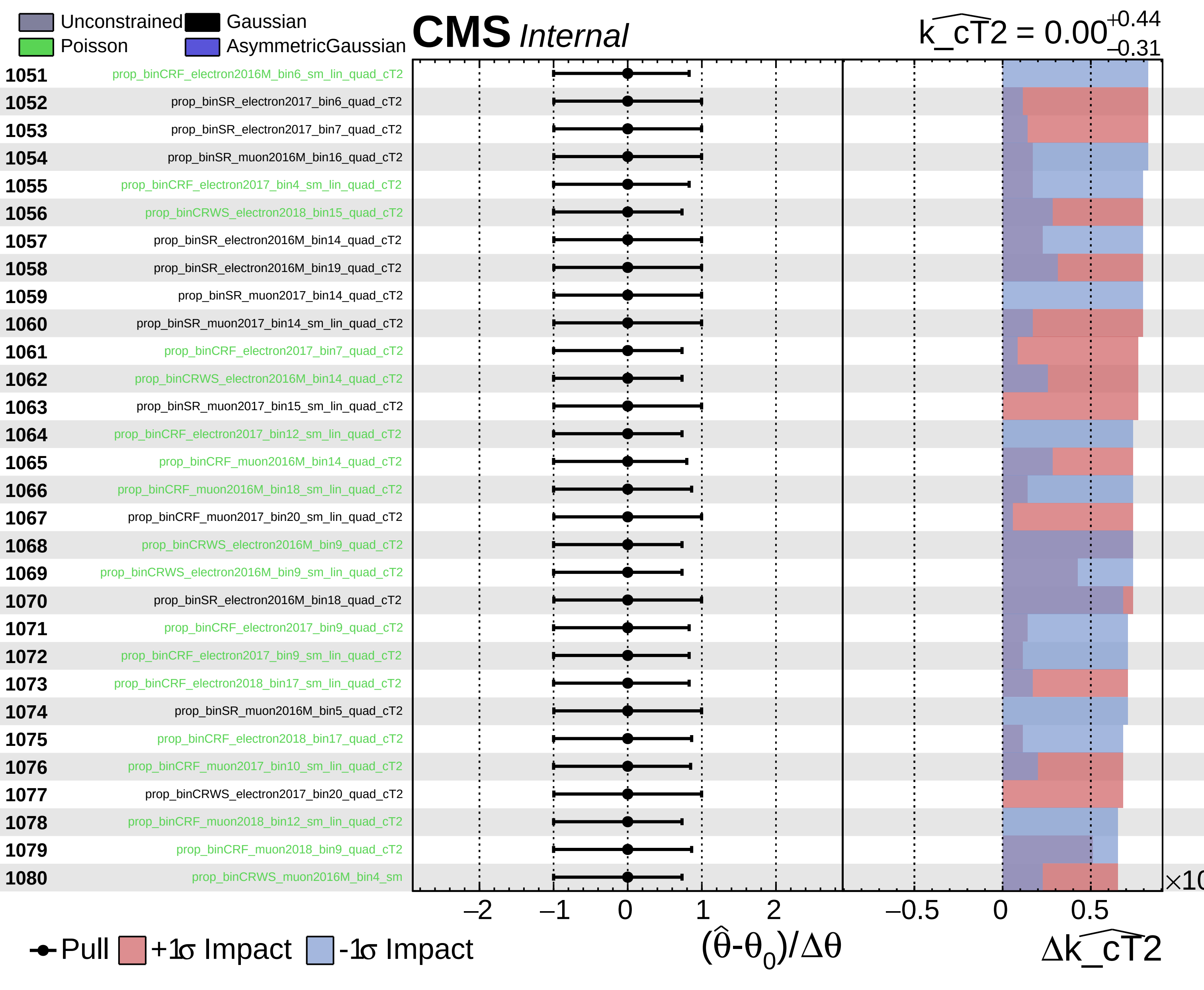
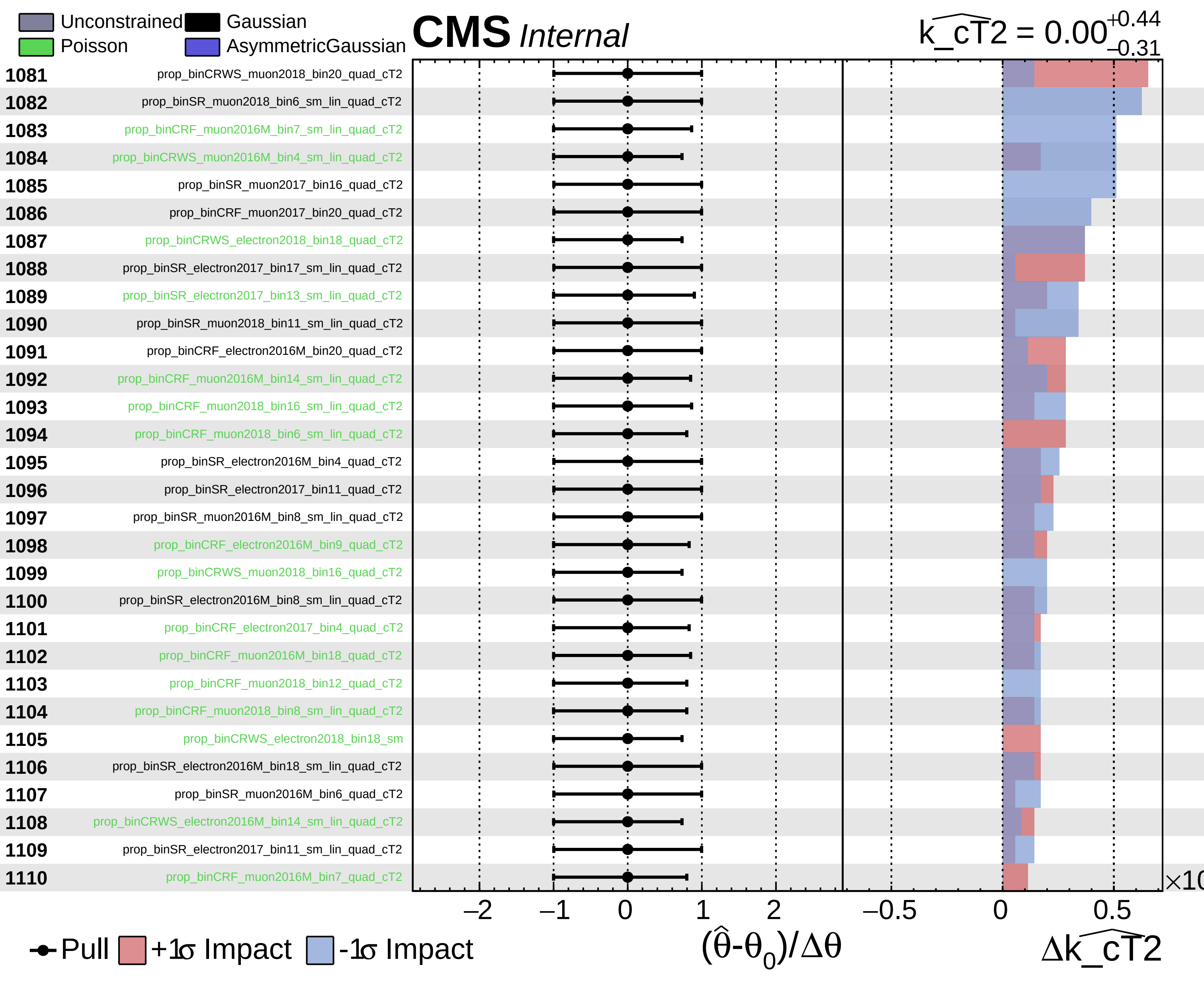
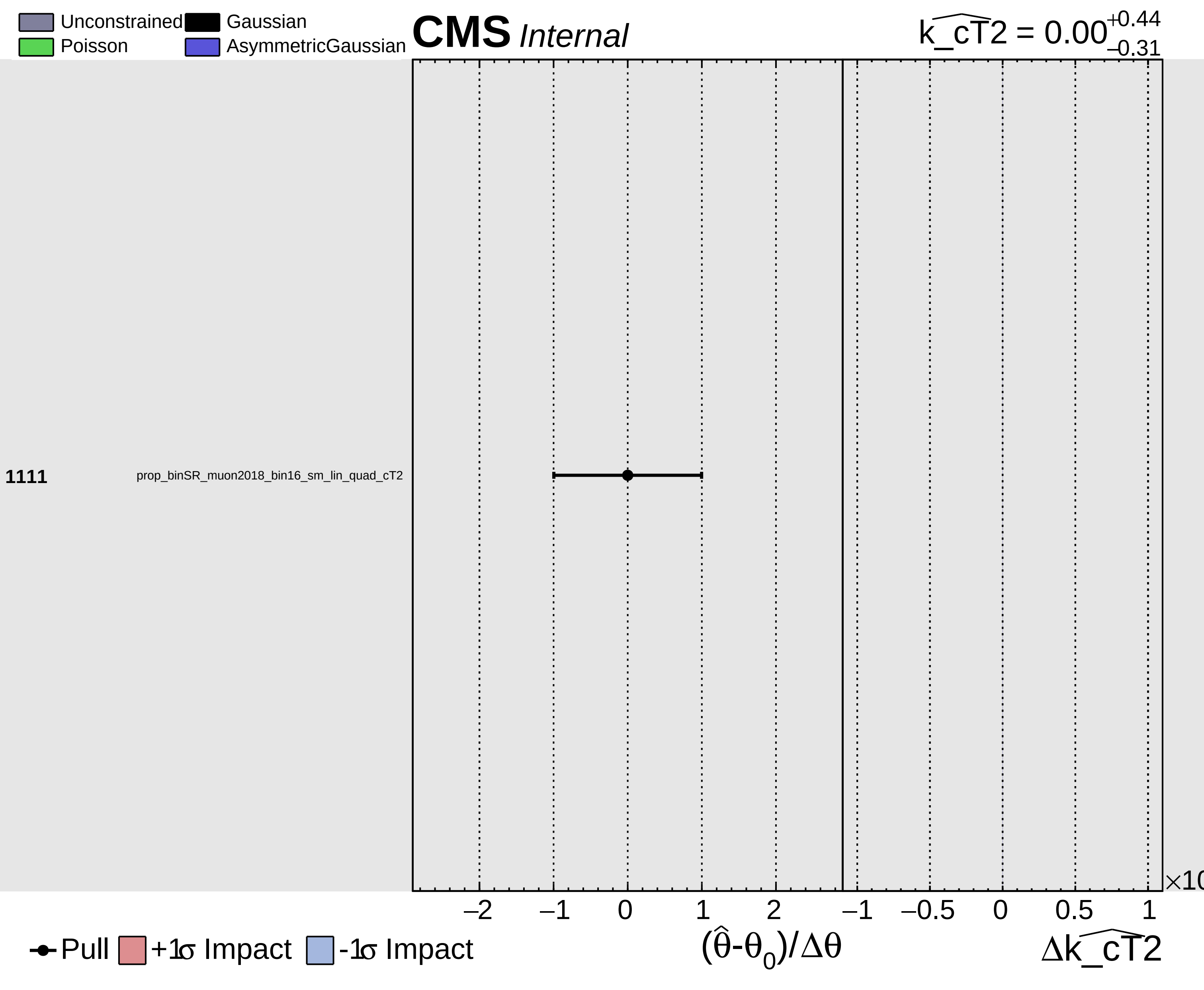
1000 prop_binCRF_electron2016M_bin18_sm_lin_quad_cT2
1001 prop_binCRWS_muon2018_bin16_sm
1002 prop_binSR_electron2017_bin6_sm_lin_quad_cT2
1003 prop_binSR_muon2018_bin14_quad_cT2
1004 prop_binCRF_electron2016M_bin9_sm_lin_quad_cT2
1005 prop_binCRF_electron2018_bin11_quad_cT2
1006 prop_binCRF_electron2016M_bin10_sm_lin_quad_cT2
1007 prop_binCRF_muon2018_bin17_quad_cT2
1008 prop_binSR_electron2018_bin18_sm_lin_quad_cT2
1009 prop_binCRWS_electron2017_bin20_sm_lin_quad_cT2
1010 prop_binCRWS_electron2018_bin15_sm_lin_quad_cT2
1011 prop_binSR_muon2016M_bin16_sm_lin_quad_cT2
1012 prop_binCRWS_muon2017_bin20_quad_cT2
1013 prop_binCRF_muon2017_bin7_quad_cT2
1014 prop_binCRF_electron2017_bin12_quad_cT2
1015 prop_binSR_muon2018_bin14_sm_lin_quad_cT2
1016 prop_binSR_muon2016M_bin19_sm_lin_quad_cT2
1017 prop_binSR_electron2016M_bin4_sm_lin_quad_cT2
1018 prop_binCRF_muon2018_bin6_quad_cT2
1019 prop_binCRF_electron2018_bin16_sm_lin_quad_cT2
1020 prop_binSR_muon2018_bin6_quad_cT2
−2 −1 0 1 2
−20 0 20
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
1021 prop_binSR_muon2018_bin11_quad_cT2
1022 prop_binSR_electron2016M_bin7_sm_lin_quad_cT2
1023 prop_binCRF_muon2018_bin8_quad_cT2
1024 prop_binCRF_muon2016M_bin20_sm_lin_quad_cT2
1025 prop_binCRF_muon2018_bin16_quad_cT2
1026 prop_binSR_muon2018_bin16_quad_cT2
1027 prop_binSR_muon2017_bin12_sm_lin_quad_cT2
1028 prop_binSR_muon2016M_bin5_sm_lin_quad_cT2
1029 prop_binSR_electron2016M_bin7_quad_cT2
1030 prop_binCRF_electron2017_bin7_sm_lin_quad_cT2
1031 prop_binSR_muon2017_bin12_quad_cT2
1032 VBS_2016M
1033 VBS_2017
1034 VBS_2018
1035 r
1.0 +0 −0
1036 tau_vsele_2016M
1037 tau_vsele_2017
1038 tau_vsele_2018
1039 tau_vsmu_2016M
1040 tau_vsmu_2017
1041 tau_vsmu_2018
1042 prop_binCRF_muon2017_bin10_quad_cT2
1043 prop_binSR_muon2016M_bin6_sm_lin_quad_cT2
1044 prop_binCRF_electron2016M_bin12_quad_cT2
1045 prop_binCRF_electron2016M_bin6_quad_cT2
1046 prop_binCRF_electron2017_bin14_quad_cT2
1047 prop_binSR_electron2017_bin15_sm_lin_quad_cT2
1048 prop_binCRF_electron2018_bin15_sm_lin_quad_cT2
1049 prop_binSR_electron2017_bin9_sm_lin_quad_cT2
1050 prop_binCRF_electron2016M_bin20_sm_lin_quad_cT2
−2 −1 0 1 2
−20 0 20
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.00^{+0.44}_{-0.31}$
1051 prop_binCRF_electron2016M_bin6_sm_lin_quad_cT2
1052 prop_binSR_electron2017_bin6_quad_cT2
1053 prop_binSR_electron2017_bin7_quad_cT2
1054 prop_binSR_muon2016M_bin16_quad_cT2
1055 prop_binCRF_electron2017_bin4_sm_lin_quad_cT2
1056 prop_binCRWS_electron2018_bin15_quad_cT2
1057 prop_binSR_electron2016M_bin14_quad_cT2
1058 prop_binSR_electron2016M_bin19_quad_cT2
1059 prop_binSR_muon2017_bin14_quad_cT2
1060 prop_binSR_muon2017_bin14_sm_lin_quad_cT2
1061 prop_binCRF_electron2017_bin7_quad_cT2
1062 prop_binCRWS_electron2016M_bin14_quad_cT2
1063 prop_binSR_muon2017_bin15_sm_lin_quad_cT2
1064 prop_binCRF_electron2017_bin12_sm_lin_quad_cT2
1065 prop_binCRF_muon2016M_bin14_quad_cT2
1066 prop_binCRF_muon2016M_bin18_sm_lin_quad_cT2
1067 prop_binCRF_muon2017_bin20_sm_lin_quad_cT2
1068 prop_binCRWS_electron2016M_bin9_quad_cT2
1069 prop_binCRWS_electron2016M_bin9_sm_lin_quad_cT2
1070 prop_binSR_electron2016M_bin18_quad_cT2
1071 prop_binCRF_electron2017_bin9_quad_cT2
1072 prop_binCRF_electron2017_bin9_sm_lin_quad_cT2
1073 prop_binCRF_electron2018_bin17_sm_lin_quad_cT2
1074 prop_binSR_muon2016M_bin5_quad_cT2
1075 prop_binCRF_electron2018_bin17_quad_cT2
1076 prop_binCRF_muon2017_bin10_sm_lin_quad_cT2
1077 prop_binCRWS_electron2017_bin20_quad_cT2
1078 prop_binCRF_muon2018_bin12_sm_lin_quad_cT2
1079 prop_binCRF_muon2018_bin9_quad_cT2
1080 prop_binCRWS_muon2016M_bin4_sm
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
1081 prop_binCRWS_muon2018_bin20_quad_cT2
1082 prop_binSR_muon2018_bin6_sm_lin_quad_cT2
1083 prop_binCRF_muon2016M_bin7_sm_lin_quad_cT2
1084 prop_binCRWS_muon2016M_bin4_sm_lin_quad_cT2
1085 prop_binSR_muon2017_bin16_quad_cT2
1086 prop_binCRF_muon2017_bin20_quad_cT2
1087 prop_binCRWS_electron2018_bin18_quad_cT2
1088 prop_binSR_electron2017_bin17_sm_lin_quad_cT2
1089 prop_binSR_electron2017_bin13_sm_lin_quad_cT2
1090 prop_binSR_muon2018_bin11_sm_lin_quad_cT2
1091 prop_binCRF_electron2016M_bin20_quad_cT2
1092 prop_binCRF_muon2016M_bin14_sm_lin_quad_cT2
1093 prop_binCRF_muon2018_bin16_sm_lin_quad_cT2
1094 prop_binCRF_muon2018_bin6_sm_lin_quad_cT2
1095 prop_binSR_electron2016M_bin4_quad_cT2
1096 prop_binSR_electron2017_bin11_quad_cT2
1097 prop_binSR_muon2016M_bin8_sm_lin_quad_cT2
1098 prop_binCRF_electron2016M_bin9_quad_cT2
1099 prop_binCRWS_muon2018_bin16_quad_cT2
1100 prop_binSR_electron2016M_bin8_sm_lin_quad_cT2
1101 prop_binCRF_electron2017_bin4_quad_cT2
1102 prop_binCRF_muon2016M_bin18_quad_cT2
1103 prop_binCRF_muon2018_bin12_quad_cT2
1104 prop_binCRF_muon2018_bin8_sm_lin_quad_cT2
1105 prop_binCRWS_electron2018_bin18_sm
1106 prop_binSR_electron2016M_bin18_sm_lin_quad_cT2
1107 prop_binSR_muon2016M_bin6_quad_cT2
1108 prop_binCRWS_electron2016M_bin14_sm_lin_quad_cT2
1109 prop_binSR_electron2017_bin11_sm_lin_quad_cT2
1110 prop_binCRF_muon2016M_bin7_quad_cT2
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.00 +0.44 −0.31
1111
prop_binSR_muon2018_bin16_sm_lin_quad_cT2
×10
−2
−1
0
1
2
−1
−0.5
0
0.5
1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2